

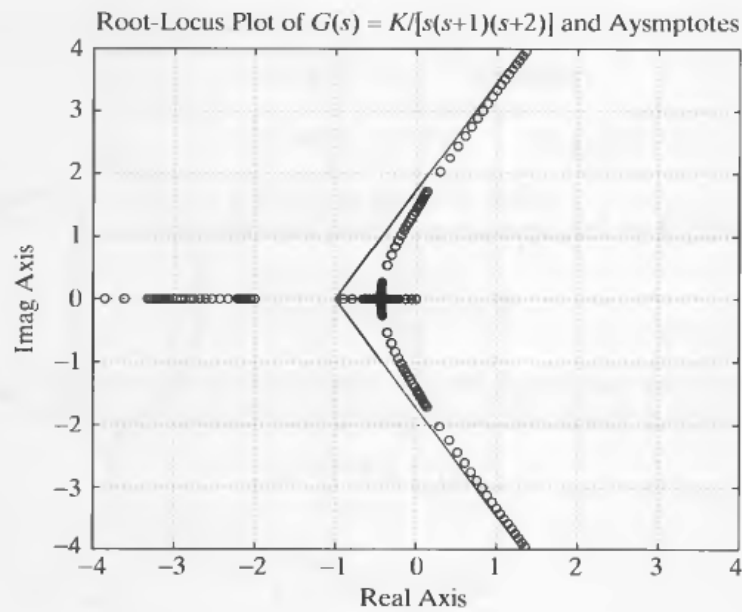


Figure 6-62
Root-locus plot.

MATLAB Program 6-14

```
% ----- Root-Locus Plot -----

num = [0 0 1 4 4];
den = [1 10 29 40 100];
r = rlocus(num,den);
plot(r,'o')
hold
Current plot held
plot(r,'-')
v = [-8 4 -6 6]; axis(v); axis('square')
grid
title('Root-Locus Plot of  $G(s) = (s + 2)^2/[(s^2 + 4)(s + 5)^2]$ ')
xlabel('Real Axis')
ylabel('Imag Axis')
```

- A-6-18.** Consider the system with transport lag shown in Figure 6-64(a). Sketch the root loci and find the two pairs of closed-loop poles nearest the $j\omega$ axis.

Using only the dominant closed-loop poles, obtain the unit-step response and sketch the response curve.

Solution. The characteristic equation is

$$\frac{2e^{-0.3s}}{s+1} + 1 = 0$$

which is equivalent to the following angle and magnitude conditions:

$$\left| \frac{2e^{-0.3s}}{s+1} \right| = \pm 180^\circ(2k+1)$$

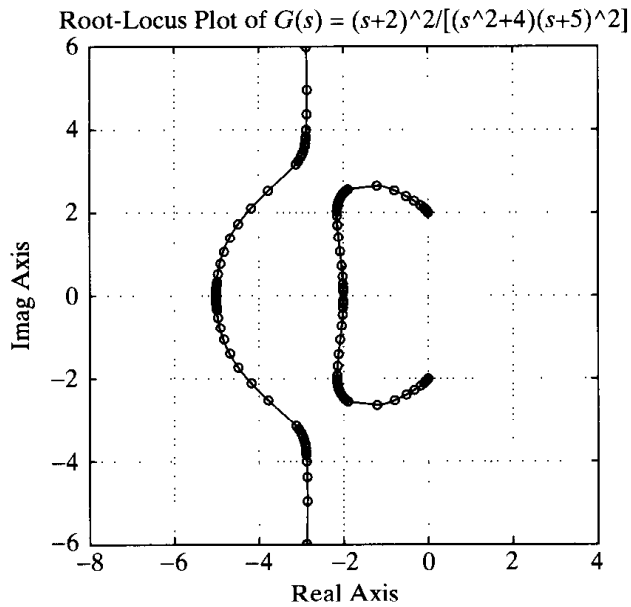


Figure 6–63
Root-locus plot.

$$\left| \frac{2e^{-0.3s}}{s+1} \right| = 1$$

The angle condition reduces to

$$\angle s+1 = \pm\pi(2k+1) - 0.3\omega \quad (\text{radians})$$

For $k = 0$,

$$\begin{aligned} \angle s+1 &= \pm\pi - 0.3\omega \quad (\text{radians}) \\ &= \pm 180^\circ - 17.2^\circ\omega \quad (\text{degrees}) \end{aligned}$$

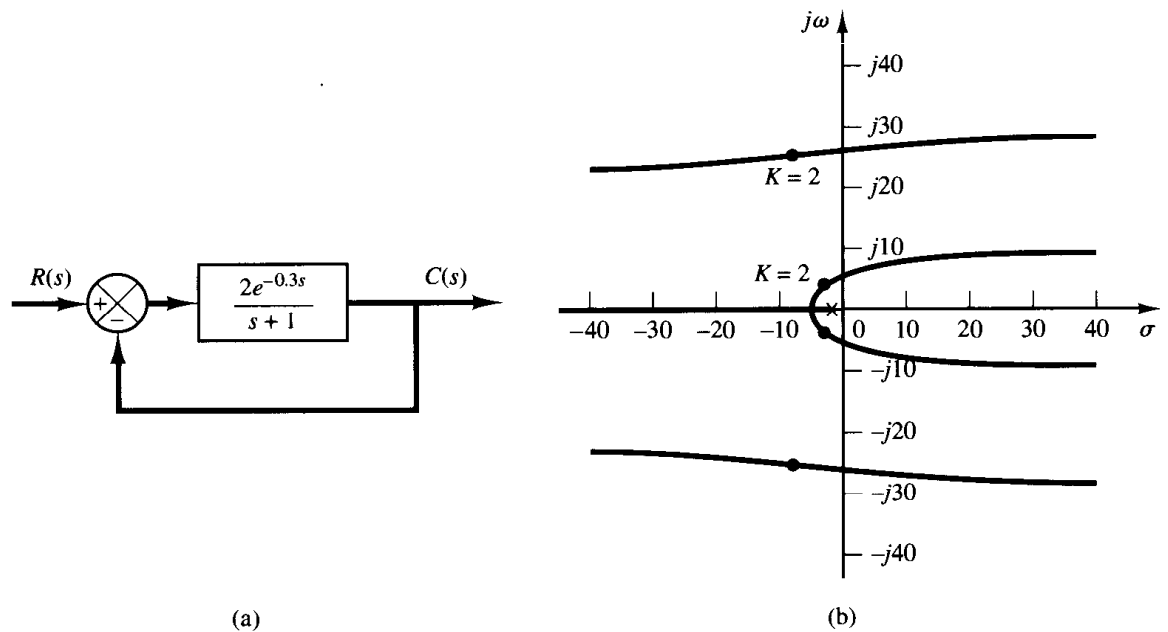


Figure 6–64
(a) Control system
with transport lag;
(b) root-locus plot.

For $k = 1$,

$$\begin{aligned}\angle s + 1 &= \pm 3\pi - 0.3\omega \quad (\text{radians}) \\ &= \pm 540^\circ - 17.2^\circ\omega \quad (\text{degrees})\end{aligned}$$

The root-locus plot for this system is shown in Figure 6-64(b).

Let us set $s = \sigma + j\omega$ in the magnitude condition and replace 2 by K . Then we obtain

$$\frac{\sqrt{(1 + \sigma)^2 + \omega^2}}{e^{-0.3\sigma}} = K$$

By evaluating K at different points on the root loci, the points may be found for which $K = 2$. These points are closed-loop poles. The dominant pair of closed-loop poles is

$$s = -2.5 \pm j3.9$$

The next pair of closed-loop poles is

$$s = -8.6 \pm j25.1$$

Using only the pair of dominant closed-loop poles, the closed-loop transfer function may be approximated as follows: Noting that

$$\begin{aligned}\frac{C(s)}{R(s)} &= \frac{2e^{-0.3s}}{1 + s + 2e^{-0.3s}} \\ &= \frac{2e^{-0.3s}}{1 + s + 2\left(1 - 0.3s + \frac{0.09s^2}{2} + \cdots\right)} \\ &= \frac{2e^{-0.3s}}{3 + 0.4s + 0.09s^2 + \cdots}\end{aligned}$$

and

$$(s + 2.5 + j3.9)(s + 2.5 - j3.9) = s^2 + 5s + 21.46$$

we may approximate $C(s)/R(s)$ by

$$\frac{C(s)}{R(s)} = \frac{\frac{2}{3}(21.46)e^{-0.3s}}{s^2 + 5s + 21.46}$$

or

$$\frac{C(s)}{R(s)} = \frac{14.31e^{-0.3s}}{(s + 2.5)^2 + 3.9^2}$$

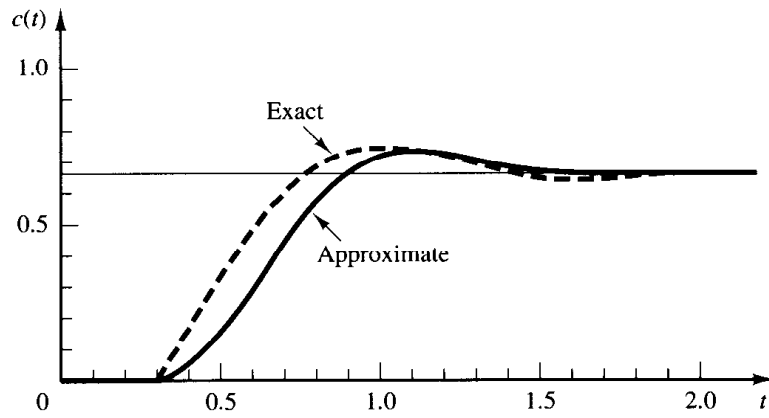
For a unit-step input,

$$C(s) = \frac{14.31e^{-0.3s}}{[(s + 2.5)^2 + 3.9^2]s}$$

Note that

$$\frac{14.31}{[(s + 2.5)^2 + 3.9^2]s} = \frac{\frac{2}{3}}{s} + \frac{-\frac{2}{3}s - \frac{10}{3}}{(s + 2.5)^2 + 3.9^2}$$

Figure 6–65
Unit-step response curves for the system shown in Figure 6–64(a).



Hence,

$$C(s) = \left(\frac{2}{3}\right)e^{-0.3s} + \left[\frac{-\frac{2}{3}s - \frac{10}{3}}{(s + 2.5)^2 + 3.9^2}\right]e^{-0.3s}$$

The inverse Laplace transform of $C(s)$ gives

$$c(t) = \frac{2}{3}[1 - e^{-2.5(t-0.3)} \cos 3.9(t - 0.3) - 0.641e^{-2.5(t-0.3)} \sin 3.9(t - 0.3)]1(t - 0.3)$$

where $1(t - 0.3)$ is the unit-step function occurring at $t = 0.3$.

Figure 6–65 shows the approximate response curve thus obtained, together with the exact unit-step response curve obtained by a computer simulation. Note that in this system a fairly good approximation can be obtained by use of only the dominant closed-loop poles.

PROBLEMS

B-6-1. Plot the root loci for the closed-loop control system with

$$G(s) = \frac{K}{s(s + 1)(s^2 + 4s + 5)}, \quad H(s) = 1$$

B-6-2. Plot the root loci for a closed-loop control system with

$$G(s) = \frac{K(s + 9)}{s(s^2 + 4s + 11)}, \quad H(s) = 1$$

Locate the closed-loop poles on the root loci such that the dominant closed-loop poles have a damping ratio equal to 0.5. Determine the corresponding value of gain K .

B-6-3. Plot the root loci for the system with

$$G(s) = \frac{K}{s(s + 0.5)(s^2 + 0.6s + 10)}, \quad H(s) = 1$$

B-6-4. Plot the root loci for a system with

$$G(s) = \frac{K}{(s^2 + 2s + 2)(s^2 + 2s + 5)}, \quad H(s) = 1$$

Determine the exact points where the root loci cross the $j\omega$ axis.

B-6-5. Show that the root loci for a control system with

$$G(s) = \frac{K(s^2 + 6s + 10)}{s^2 + 2s + 10}, \quad H(s) = 1$$

are arcs of the circle centered at the origin with radius equal to $\sqrt{10}$.

B-6-6. Plot the root loci for a closed-loop control system with

$$G(s) = \frac{K(s + 0.2)}{s^2(s + 3.6)}, \quad H(s) = 1$$

B-6-7. Plot the root loci for a closed-loop control system with

$$G(s) = \frac{K(s + 0.5)}{s^3 + s^2 + 1}, \quad H(s) = 1$$

B-6-8. Plot the root loci for the system shown in Figure 6-66. Determine the range of gain K for stability.

B-6-9. Consider a unity-feedback control system with the following feedforward transfer function:

$$G(s) = \frac{K}{s(s^2 + 4s + 8)}$$

Plot the root loci for the system. If the value of gain K is set equal to 2, where are the closed-loop poles located?

B-6-10. Consider the system shown in Figure 6-67. Determine the values of the gain K and the velocity feedback coefficient K_h so that the closed-loop poles are at $s = -1 \pm j\sqrt{3}$. Then, using the determined value of K_h , plot the root loci.

B-6-11. Consider the system shown in Figure 6-68. The system involves velocity feedback. Determine the value of gain K such that the dominant closed-loop poles have a damping ratio of 0.5. Using the gain K thus determined, obtain the unit-step response of the system.

B-6-12. Consider the system whose open-loop transfer function $G(s)H(s)$ is given by

$$G(s)H(s) = \frac{K}{(s^2 + 2s + 2)(s^2 + 2s + 5)} \\ = \frac{K}{s^4 + 4s^3 + 11s^2 + 14s + 10}$$

Plot a root-locus diagram with MATLAB.

B-6-13. Consider the system whose open-loop transfer function is given by

$$G(s)H(s) = \frac{K(s - 0.6667)}{s^4 + 3.3401s^3 + 7.0325s^2}$$

Show that the equation for the asymptotes is given by

$$G_a(s)H_a(s) = \frac{K}{s^3 + 4.0068s^2 + 5.3515s + 2.3825}$$

Using MATLAB, plot the root loci and asymptotes for the system.

B-6-14. Consider the unity-feedback system whose feedforward transfer function is

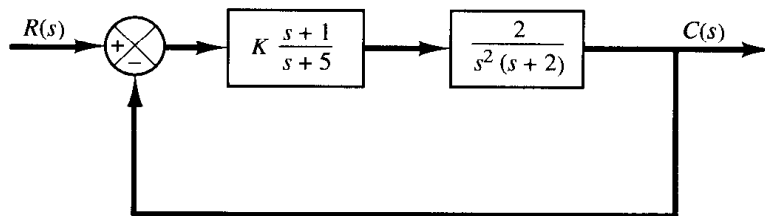


Figure 6-66
Control system.

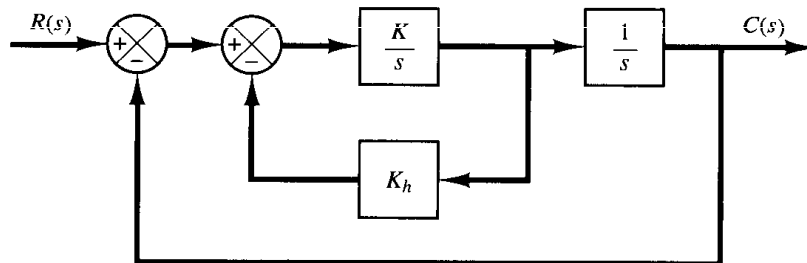


Figure 6-67
Control system.

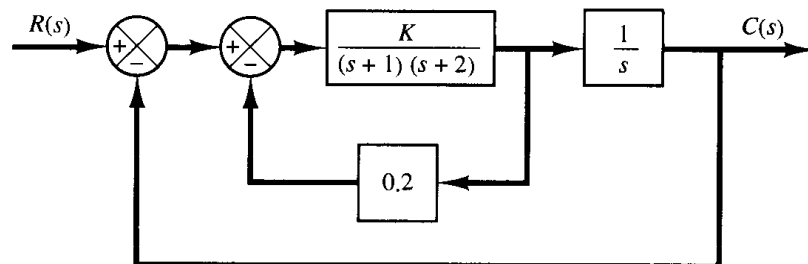


Figure 6-68
Control system.

$$G(s) = \frac{K}{s(s+1)}$$

The constant-gain locus for the system for a given value of K is defined by the following equation:

$$\left| \frac{K}{s(s+1)} \right| = 1$$

Show that the constant-gain loci for $0 \leq K \leq \infty$ may be given by

$$[\sigma(\sigma+1) + \omega^2]^2 + \omega^2 = K^2$$

Sketch the constant-gain loci for $K = 1, 2, 5, 10$, and 20 on the s plane.

B-6-15. Consider the system shown in Figure 6-69. Plot the root loci. Locate the closed-loop poles when the gain K is set equal to 2.

B-6-16. Consider the system shown in Figure 6-70. Plot the root loci as a varies from 0 to ∞ . Determine the value of a such that the damping ratio of the dominant closed-loop poles is 0.5.

B-6-17. Consider the system shown in Figure 6-71. Plot the root loci as the value of k varies from 0 to ∞ . What value of k will give the damping ratio of the dominant closed-loop poles equal to 0.5? Find the static velocity error constant with this value of k .

B-6-18. Plot the root loci for the system shown in Figure 6-72. Show that the system may become unstable for large values of K .

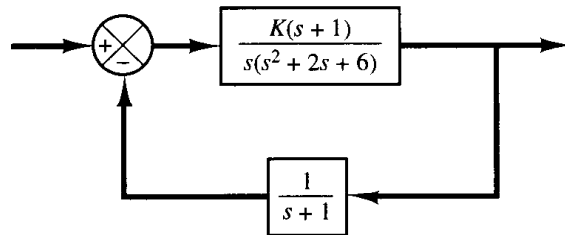


Figure 6-69
Control system.

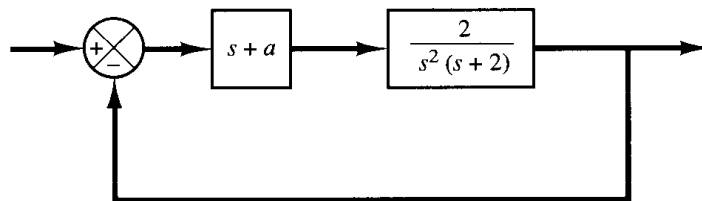


Figure 6-70
Control system.

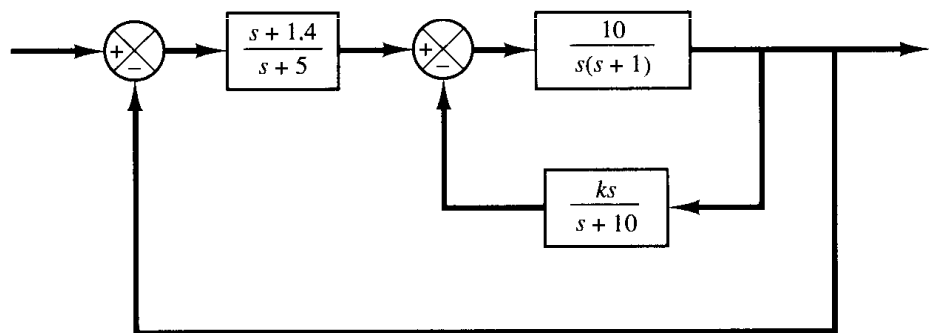


Figure 6-71
Control system.

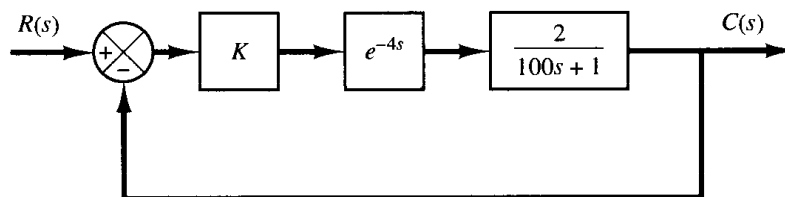


Figure 6-72
Control system.

B-6-19. Plot the root contours for the system shown in Figure 6-73 when the gain K and parameter a vary, respectively, from zero to infinity.

B-6-20. Consider the system shown in Figure 6-74. Assuming that the value of gain K varies from 0 to ∞ , plot the

root loci when $K_h = 0.5$. Then sketch the root contours for $0 \leq K < \infty$ and $0 \leq K_h < \infty$. Locate the closed-loop poles on the root contour when $K = 10$ and $K_h = 0.5$.

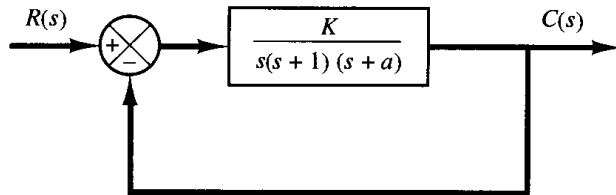


Figure 6-73
Control system.

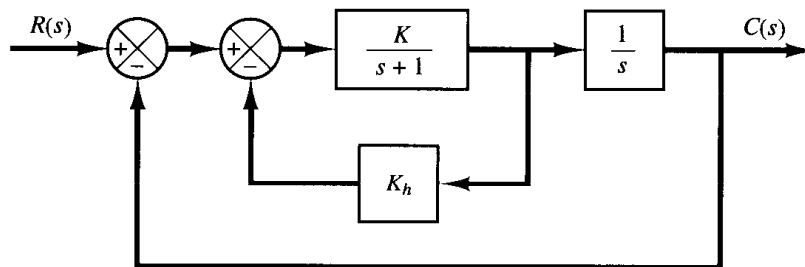


Figure 6-74
Control system.

7

Control Systems Design by the Root- Locus Method

7-1 INTRODUCTION

The primary objective of this chapter is to present procedures for the design and compensation of single-input–single-output linear time-invariant control systems. Compensation is the modification of the system dynamics to satisfy the given specifications. The approach to the control system design and compensation used in this chapter is the root-locus approach. (The frequency-response approach and the state-space approach to the control systems design and compensation will be presented in Chapter 9 and Chapter 11, respectively.)

Performance specifications. Control systems are designed to perform specific tasks. The requirements imposed on the control system are usually spelled out as performance specifications. They generally relate to accuracy, relative stability, and speed of response.

For routine design problems, the performance specifications may be given in terms of precise numerical values. In other cases, they may be given partially in terms of precise numerical values and partially in terms of qualitative statements. In the latter case, the specifications may have to be modified during the course of design since the given specifications may never be satisfied (because of conflicting requirements) or may lead to a very expensive system.

Generally, the performance specifications should not be more stringent than necessary to perform the given task. If the accuracy at steady-state operation is of prime importance in a given control system, then we should not require unnecessarily rigid performance specifications on the transient response since such specifications will re-

quire expensive components. Remember that the most important part of control system design is to state the performance specifications precisely so that they will yield an optimal control system for the given purpose.

System compensation. Setting the gain is the first step in adjusting the system for satisfactory performance. In many practical cases, however, the adjustment of the gain alone may not provide sufficient alteration of the system behavior to meet the given specifications. As is frequently the case, increasing the gain value will improve the steady-state behavior but will result in poor stability or even instability. It is then necessary to redesign the system (by modifying the structure or by incorporating additional devices or components) to alter the overall behavior so that the system will behave as desired. Such a redesign or addition of a suitable device is called *compensation*. A device inserted into the system for the purpose of satisfying the specifications is called a *compensator*. The compensator compensates for deficit performance of the original system.

Series compensation and feedback (or parallel) compensation. Figures 7–1 (a) and (b) show compensation schemes commonly used for feedback control systems. Figure 7–1(a) shows the configuration where the compensator $G_c(s)$ is placed in series with the plant. This scheme is called *series compensation*.

An alternative to series compensation is to feed back the signal(s) from some element(s) and place a compensator in the resulting inner feedback path, as shown in Figure 7–1(b). Such compensation is called *feedback compensation* or *parallel compensation*.

In compensating control systems, we see that the problem usually boils down to a suitable design of a series or feedback compensator. The choice between series compensation and feedback compensation depends on the nature of the signals in the

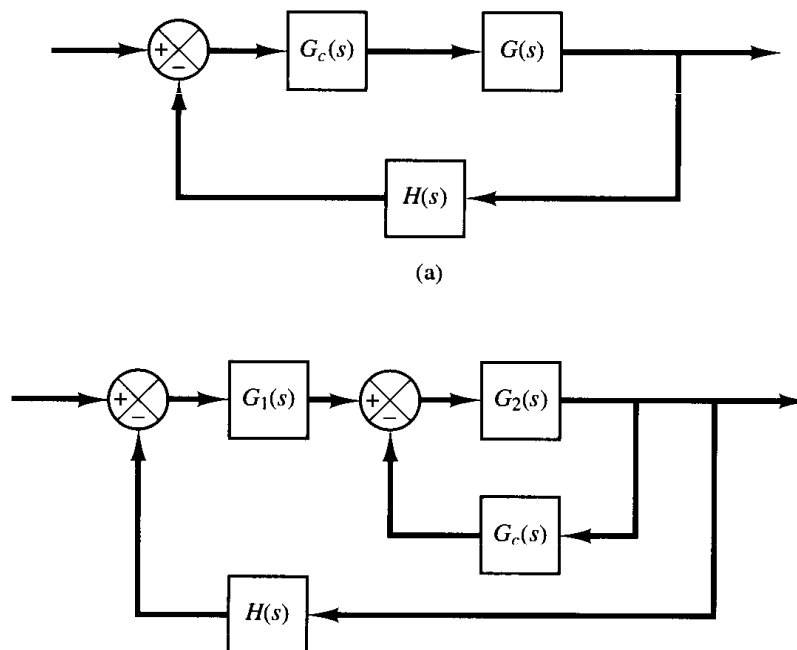


Figure 7–1
(a) Series compensation; (b) feedback or parallel compensation.

system, the power levels at various points, available components, the designer's experience, economic considerations, and so on.

In general, series compensation may be simpler than feedback compensation; however, series compensation frequently requires additional amplifiers to increase the gain and/or to provide isolation. (To avoid power dissipation, the series compensator is inserted at the lowest energy point in the feedforward path.) Note that, in general, the number of components required in feedback compensation will be less than the number of components in series compensation, provided a suitable signal is available, because the energy transfer is from a higher power level to a lower level. (This means that additional amplifiers may not be necessary.)

In discussing compensators, we frequently use such terminologies as *lead network*, *lag network*, and *lag-lead network*. As stated in Section 5-9, if a sinusoidal input e_i is applied to the input of a network and the steady-state output e_o (which is also sinusoidal) has a phase lead, then the network is called a lead network. (The amount of phase lead angle is a function of the input frequency.) If the steady-state output e_o has a phase lag, then the network is called a lag network. In a lag-lead network, both phase lag and phase lead occur in the output but in different frequency regions; phase lag occurs in the low-frequency region and phase lead occurs in the high-frequency region. A compensator having a characteristic of a lead network, lag network, or lag-lead network is called a lead compensator, lag compensator, or lag-lead compensator.

Compensators. If a compensator is needed to meet the performance specifications, the designer must realize a physical device that has the prescribed transfer function of the compensator.

Numerous physical devices have been used for such purposes. In fact, many noble and useful ideas for physically constructing compensators may be found in the literature.

Among the many kinds of compensators, widely employed compensators are the lead compensators, lag compensators, lag-lead compensators, and velocity-feedback (tachometer) compensators. In this chapter we shall limit our discussions mostly to these types. Lead, lag, and lag-lead compensators may be electronic devices (such as circuits using operational amplifiers) or RC networks (electrical, mechanical, pneumatic, hydraulic, or combinations thereof) and amplifiers.

In the actual design of a control system, whether to use an electronic, pneumatic, or hydraulic compensator is a matter that must be decided partially based on the nature of the controlled plant. For example, if the controlled plant involves flammable fluid, then we have to choose pneumatic components (both a compensator and an actuator) to avoid the possibility of sparks. If, however, no fire hazard exists, then electronic compensators are most commonly used. (In fact, we often transform nonelectrical signals into electrical signals because of the simplicity of transmission, increased accuracy, increased reliability, ease of compensation, and the like.)

Design procedures. In the trial-and-error approach to system design, we set up a mathematical model of the control system and adjust the parameters of a compensator. The most time-consuming part of such work is the checking of the system performance by analysis with each adjustment of the parameters. The designer should make use of a digital computer to avoid much of the numerical drudgery necessary for this checking.

Once a satisfactory mathematical model has been obtained, the designer must construct a prototype and test the open-loop system. If absolute stability of the closed loop is assured, the designer closes the loop and tests the performance of the resulting closed-loop system. Because of the neglected loading effects among the components, nonlinearities, distributed parameters, and so on, which were not taken into consideration in the original design work, the actual performance of the prototype system will probably differ from the theoretical predictions. Thus the first design may not satisfy all the requirements on performance. By trial and error, the designer must make changes in the prototype until the system meets the specifications. In doing this, he or she must analyze each trial, and the results of the analysis must be incorporated into the next trial. The designer must see that the final system meets the performance specifications and, at the same time, is reliable and economical.

It is noted that in designing control systems by the root-locus or frequency-response methods the final result is not unique, because the best or optimal solution may not be precisely defined if the time-domain specifications or frequency-domain specifications are given.

Outline of the chapter. Section 7-1 has presented an introduction to the compensation of control systems. Section 7-2 discusses preliminary considerations for the root-locus approach to the control systems design. Section 7-3 treats details of the lead compensation techniques based on the root-locus method. Section 7-4 deals with the lag compensation techniques by the root-locus method. Section 7-5 presents lag-lead compensation techniques. Detailed discussions of the design of lag-lead compensators are presented.

7-2 PRELIMINARY DESIGN CONSIDERATIONS

In building a control system, we know that proper modification of the plant dynamics may be a simple way to meet the performance specifications. This, however, may not be possible in many practical situations because the plant may be fixed and may not be modified. Then we must adjust parameters other than those in the fixed plant. In this book, we assume that the plant is given and unalterable.

The design problems, therefore, become those of improving system performance by insertion of a compensator. Compensation of a control system is reduced to the design of a filter whose characteristics tend to compensate for the undesirable and unalterable characteristics of the plant. Our discussions are limited to continuous-time compensators.

In Section 7-3 through 7-5, we shall specifically consider the design of lead compensators, lag compensators, and lag-lead compensators. In such design problems, we place a compensator in series with the unalterable transfer function $G(s)$ to obtain desirable behavior. The main problem then involves the judicious choice of the pole(s) and zero(s) of the compensator $G_c(s)$ to alter the root locus (or frequency response) so that the performance specifications will be met.

Root-locus approach to control system design. The root-locus method is a graphical method for determining the locations of all closed-loop poles from knowledge

of the locations of the open-loop poles and zeros as some parameter (usually the gain) is varied from zero to infinity. The method yields a clear indication of the effects of parameter adjustment.

In practice, the root-locus plot of a system may indicate that the desired performance cannot be achieved just by the adjustment of gain. In fact, in some cases, the system may not be stable for all values of gain. Then it is necessary to reshape the root loci to meet the performance specifications.

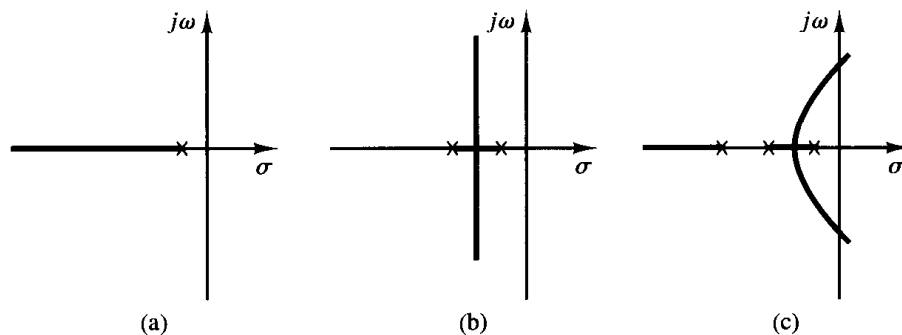
In designing a control system, if other than a gain adjustment is required, we must modify the original root loci by inserting a suitable compensator. Once the effects on the root locus of the addition of poles and/or zeros are fully understood, we can readily determine the locations of the pole(s) and zero(s) of the compensator that will reshape the root locus as desired. In essence, in the design by the root-locus method, the root loci of the system are reshaped through the use of a compensator so that a pair of dominant closed-loop poles can be placed at the desired location. (Often, the damping ratio and undamped natural frequency of a pair of dominant closed-loop poles are specified.)

Effects of the addition of poles. The addition of a pole to the open-loop transfer function has the effect of pulling the root locus to the right, tending to lower the system's relative stability and to slow down the settling of the response. (Remember that the addition of integral control adds a pole at the origin, thus making the system less stable.) Figure 7-2 shows examples of root loci illustrating the effects of the addition of a pole to a single-pole system and the addition of two poles to a single-pole system.

Effects of the addition of zeros. The addition of a zero to the open-loop transfer function has the effect of pulling the root locus to the left, tending to make the system more stable and to speed up the settling of the response. (Physically, the addition of a zero in the feedforward transfer function means the addition of derivative control to the system. The effect of such control is to introduce a degree of anticipation into the system and speed up the transient response.) Figure 7-3(a) shows the root loci for a system that is stable for small gain but unstable for large gain. Figures 7-3(b), (c), and (d) show root-locus plots for the system when a zero is added to the open-loop transfer

Figure 7-2

(a) Root-locus plot of a single-pole system; (b) root-locus plot of a two-pole system; (c) root-locus plot of a three-pole system.



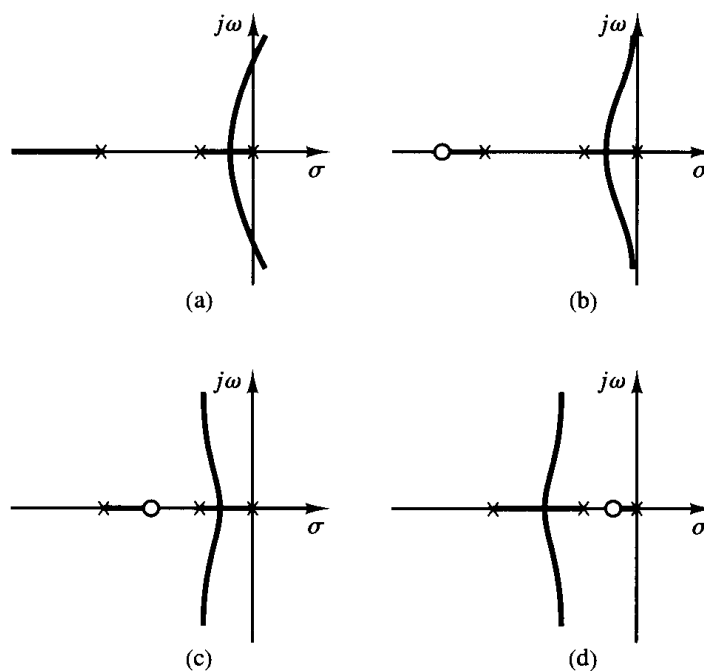


Figure 7-3
(a) Root-locus plot of a three-pole system; (b), (c), and (d) root-locus plots showing effects of addition of a zero to the three-pole system.

function. Notice that when a zero is added to the system of Figure 7-3(a) it becomes stable for all values of gain.

7-3 LEAD COMPENSATION

Lead compensators. There are many ways to realize continuous-time (or analog) lead compensators, such as electronic networks using operational amplifiers, electrical RC networks, and mechanical spring-dashpot systems. Compensators using operational amplifiers are frequently used in practice. (Refer to Chapter 5 for networks using operational amplifiers.)

Figure 7-4 shows an electronic circuit using operational amplifiers. The transfer function for this circuit was obtained in Chapter 5 as follows:

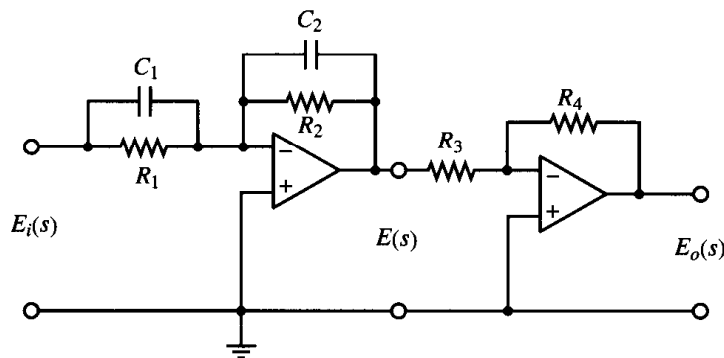


Figure 7-4
Electronic circuit that is a lead network if $R_1C_1 > R_2C_2$ and a lag network if $R_1C_1 < R_2C_2$.

$$\begin{aligned}
\frac{E_o(s)}{E_i(s)} &= \frac{R_2 R_4}{R_1 R_3} \frac{R_1 C_1 s + 1}{R_2 C_2 s + 1} = \frac{R_4 C_1}{R_3 C_2} \frac{s + \frac{1}{R_1 C_1}}{s + \frac{1}{R_2 C_2}} \\
&= K_c \alpha \frac{T s + 1}{\alpha T s + 1} = K_c \frac{s + \frac{1}{T}}{s + \frac{1}{\alpha T}} \quad (7-1)
\end{aligned}$$

where

$$T = R_1 C_1, \quad \alpha T = R_2 C_2, \quad K_c = \frac{R_4 C_1}{R_3 C_2}$$

Notice that

$$K_c \alpha = \frac{R_4 C_1}{R_3 C_2} \frac{R_2 C_2}{R_1 C_1} = \frac{R_2 R_4}{R_1 R_3}, \quad \alpha = \frac{R_2 C_2}{R_1 C_1}$$

This network has a dc gain of $K_c \alpha = R_2 R_4 / (R_1 R_3)$.

From Equation (7-1), we see that this network is a lead network if $R_1 C_1 > R_2 C_2$, or $\alpha < 1$. It is a lag network if $R_1 C_1 < R_2 C_2$. The pole-zero configurations of this network when $R_1 C_1 > R_2 C_2$ and $R_1 C_1 < R_2 C_2$ are shown in Figure 7-5(a) and (b), respectively.

Lead compensation techniques based on the root-locus approach. The root-locus approach to design is very powerful when the specifications are given in terms of time-domain quantities, such as the damping ratio and undamped natural frequency of the desired dominant closed-loop poles, maximum overshoot, rise time, and settling time.

Consider a design problem in which the original system either is unstable for all values of gain or is stable but has undesirable transient-response characteristics. In such a

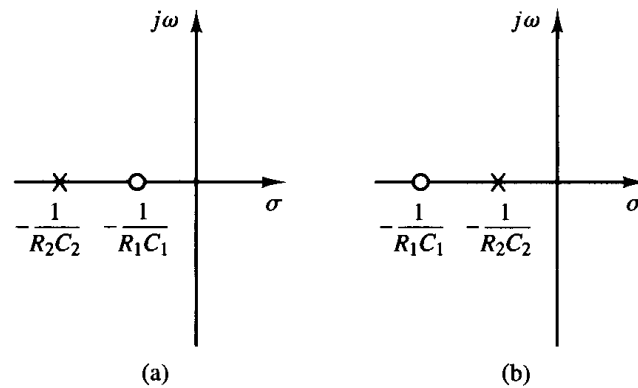


Figure 7-5
Pole-zero configurations: (a) lead network; (b) lag network.

case, the reshaping of the root locus is necessary in the broad neighborhood of the $j\omega$ axis and the origin in order that the dominant closed-loop poles be at desired locations in the complex plane. This problem may be solved by inserting an appropriate lead compensator in cascade with the feedforward transfer function.

The procedures for designing a lead compensator for the system shown in Figure 7-6 by the root-locus method may be stated as follows:

1. From the performance specifications, determine the desired location for the dominant closed-loop poles.
2. By drawing the root-locus plot, ascertain whether or not the gain adjustment alone can yield the desired closed-loop poles. If not, calculate the angle deficiency ϕ . This angle must be contributed by the lead compensator if the new root locus is to pass through the desired locations for the dominant closed-loop poles.
3. Assume the lead compensator $G_c(s)$ to be

$$G_c(s) = K_c \alpha \frac{Ts + 1}{\alpha Ts + 1} = K_c \frac{s + \frac{1}{T}}{s + \frac{1}{\alpha T}}, \quad (0 < \alpha < 1)$$

where α and T are determined from the angle deficiency. K_c is determined from the requirement of the open-loop gain.

4. If static error constants are not specified, determine the location of the pole and zero of the lead compensator so that the lead compensator will contribute the necessary angle ϕ . If no other requirements are imposed on the system, try to make the value of α as large as possible. A larger value of α generally results in a larger value of K_v , which is desirable. (If a particular static error constant is specified, it is generally simpler to use the frequency-response approach.)
5. Determine the open-loop gain of the compensated system from the magnitude condition.

Once a compensator has been designed, check to see whether all performance specifications have been met. If the compensated system does not meet the performance specifications, then repeat the design procedure by adjusting the compensator pole and zero until all such specifications are met. If a large static error constant is required, cascade a lag network or alter the lead compensator to a lag-lead compensator.

Note that if the selected dominant closed-loop poles are not really dominant, it will be necessary to modify the location of the pair of such selected dominant closed-loop

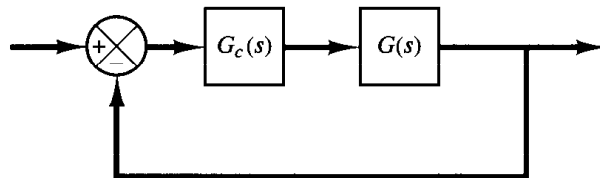


Figure 7-6
Control system.

poles. (The closed-loop poles other than dominant ones modify the response obtained from the dominant closed-loop poles alone. The amount of modification depends on the location of these remaining closed-loop poles.) Also, the closed-loop zeros affect the response if they are located near the origin.

EXAMPLE 7-1

Consider the system shown in Figure 7-7(a). The feedforward transfer function is

$$G(s) = \frac{4}{s(s+2)}$$

The root-locus plot for this system is shown in Figure 7-7(b). The closed-loop transfer function becomes

$$\begin{aligned} \frac{C(s)}{R(s)} &= \frac{4}{s^2 + 2s + 4} \\ &= \frac{4}{(s + 1 + j\sqrt{3})(s + 1 - j\sqrt{3})} \end{aligned}$$

The closed-loop poles are located at

$$s = -1 \pm j\sqrt{3}$$

The damping ratio of the closed-loop poles is 0.5. The undamped natural frequency of the closed-loop poles is 2 rad/sec. The static velocity error constant is 2 sec^{-1} .

It is desired to modify the closed-loop poles so that an undamped natural frequency $\omega_n = 4 \text{ rad/sec}$ is obtained, without changing the value of the damping ratio, $\zeta = 0.5$.

Recall that in the complex plane the damping ratio ζ of a pair of complex conjugate poles can be expressed in terms of the angle θ , which is measured from the $j\omega$ axis, as shown in Figure 7-8(a), with

$$\zeta = \sin \theta$$

In other words, lines of constant damping ratio ζ are radial lines passing through the origin as shown in Figure 7-8(b). For example, a damping ratio of 0.5 requires that the complex poles lie on the lines drawn through the origin making angles of $\pm 60^\circ$ with the negative real axis. (If the real part of a pair of complex poles is positive, which means that the system is unstable, the cor-

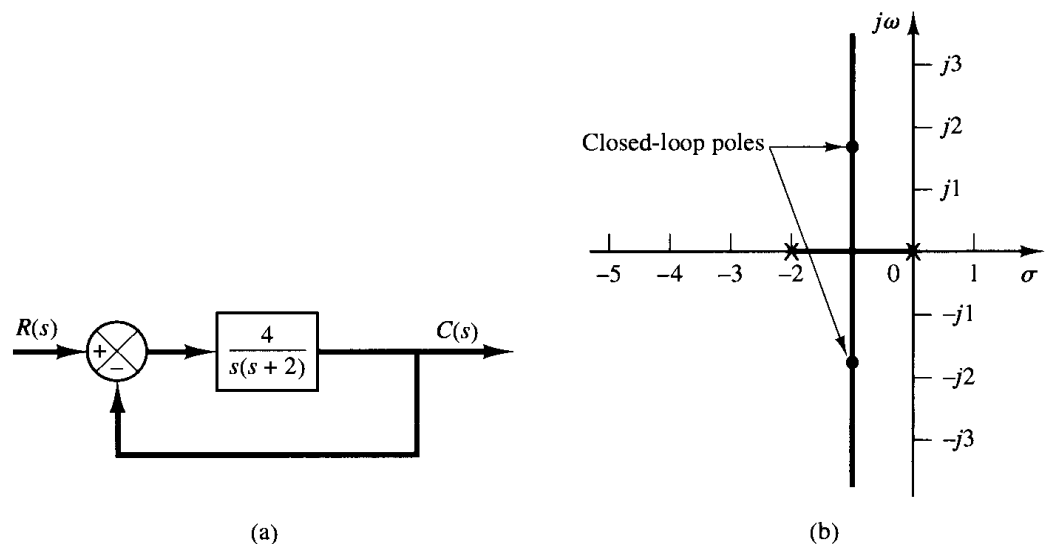


Figure 7-7
(a) Control system;
(b) root-locus plot.

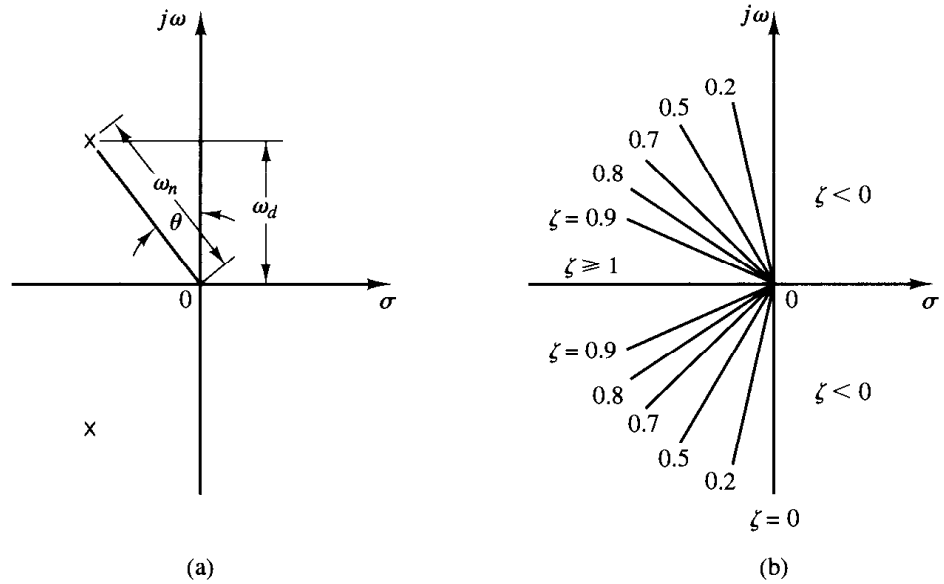


Figure 7-8
(a) Complex poles;
(b) lines of constant
damping ratio ζ .

responding ζ is negative.) The damping ratio determines the angular location of the poles, while the distance of the pole from the origin is determined by the undamped natural frequency ω_n .

In the present example, the desired locations of the closed-loop poles are

$$s = -2 \pm j2\sqrt{3}$$

In some cases, after the root loci of the original system have been obtained, the dominant closed-loop poles may be moved to the desired location by simple gain adjustment. This is, however, not the case for the present system. Therefore, we shall insert a lead compensator in the feedforward path.

A general procedure for determining the lead compensator is as follows: First, find the sum of the angles at the desired location of one of the dominant closed-loop poles with the open-loop poles and zeros of the original system, and determine the necessary angle ϕ to be added so that the total sum of the angles is equal to $\pm 180^\circ(2k + 1)$. The lead compensator must contribute this angle ϕ . (If the angle ϕ is quite large, then two or more lead networks may be needed rather than a single one.)

If the original system has the open-loop transfer function $G(s)$, then the compensated system will have the open-loop transfer function

$$G_c(s)G(s) = \left(K_c \frac{s + \frac{1}{T}}{s + \frac{1}{\alpha T}} \right) G(s)$$

where

$$G_c(s) = K_c \alpha \frac{Ts + 1}{\alpha Ts + 1} = K_c \frac{s + \frac{1}{T}}{s + \frac{1}{\alpha T}}, \quad (0 < \alpha < 1)$$

Notice that there are many possible values for T and α that will yield the necessary angle contribution at the desired closed-loop poles.

The next step is to determine the locations of the zero and pole of the lead compensator. There are many possibilities for the choice of such locations. (See the comments at the end of this example problem.) In what follows, we shall introduce a procedure to obtain the largest possible value for α . (Note that a larger value of α will produce a larger value of K_v . In most cases, the larger the K_v is, the better the system performance.) First, draw a horizontal line passing through point P , the desired location for one of the dominant closed-loop poles. This is shown as line PA in Figure 7-9. Draw also a line connecting point P and the origin. Bisect the angle between the lines PA and PO , as shown in Figure 7-9. Draw two lines PC and PD that make angles $\pm\phi/2$ with the bisector PB . The intersections of PC and PD with the negative real axis give the necessary location for the pole and zero of the lead network. The compensator thus designed will make point P a point on the root locus of the compensated system. The open-loop gain is determined by use of the magnitude condition.

In the present system, the angle of $G(s)$ at the desired closed-loop pole is

$$\left| \frac{4}{s(s+2)} \right|_{s=-2+j2\sqrt{3}} = -210^\circ$$

Thus, if we need to force the root locus to go through the desired closed-loop pole, the lead compensator must contribute $\phi = 30^\circ$ at this point. By following the foregoing design procedure, we determine the zero and pole of the lead compensator, as shown in Figure 7-10, to be

$$\text{Zero at } s = -2.9, \quad \text{Pole at } s = -5.4$$

or

$$T = \frac{1}{2.9} = 0.345, \quad \alpha T = \frac{1}{5.4} = 0.185$$

Thus $\alpha = 0.537$. The open-loop transfer function of the compensated system becomes

$$G_c(s)G(s) = K_c \frac{s+2.9}{s+5.4} \frac{4}{s(s+2)} = \frac{K(s+2.9)}{s(s+2)(s+5.4)}$$

where $K = 4K_c$. The root-locus plot for the compensated system is shown in Figure 7-10. The gain K is evaluated from the magnitude condition as follows: Referring to the root-locus plot for the compensated system shown in Figure 7-10, the gain K is evaluated from the magnitude condition as

$$\left| \frac{K(s+2.9)}{s(s+2)(s+5.4)} \right|_{s=-2+j2\sqrt{3}} = 1$$

or

$$K = 18.7$$

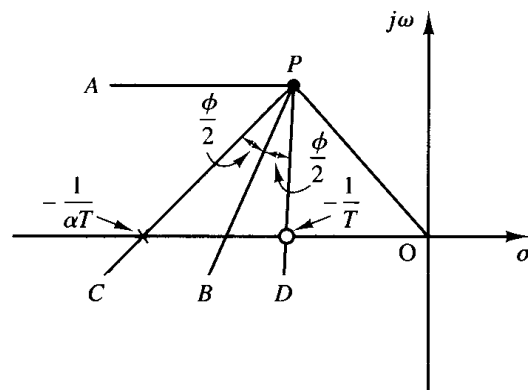


Figure 7-9
Determination of the pole and zero
of a lead network.

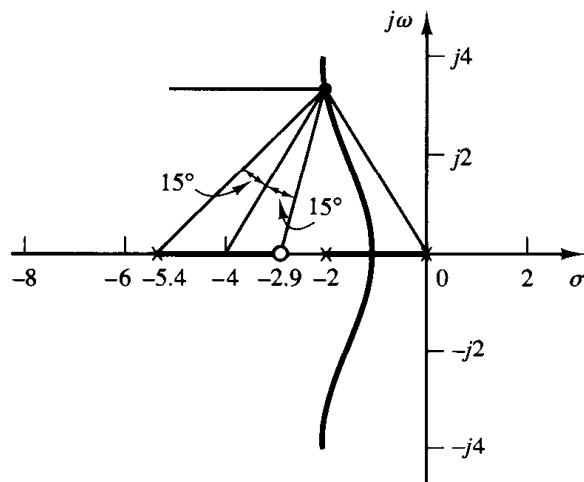


Figure 7-10
Root-locus plot of the compensated system.

It follows that

$$G_c(s)G(s) = \frac{18.7(s + 2.9)}{s(s + 2)(s + 5.4)}$$

The constant K_c of the lead compensator is

$$K_c = \frac{18.7}{4} = 4.68$$

Hence, $K_c\alpha = 2.51$. The lead compensator, therefore, has the transfer function

$$G_s(s) = 2.51 \frac{0.345s + 1}{0.185s + 1} = 4.68 \frac{s + 2.9}{s + 5.4}$$

If the electronic circuit using operational amplifiers as shown in Figure 7-4 is used as the lead compensator just designed, then the parameter values of the lead compensator are determined from

$$\frac{E_o(s)}{E_i(s)} = \frac{R_2 R_4}{R_1 R_3} \frac{R_1 C_1 s + 1}{R_2 C_2 s + 1} = 2.51 \frac{0.345s + 1}{0.185s + 1}$$

as shown in Figure 7-11, where we have arbitrarily chosen $C_1 = C_2 = 10 \mu\text{F}$ and $R_3 = 10 \text{ k}\Omega$.

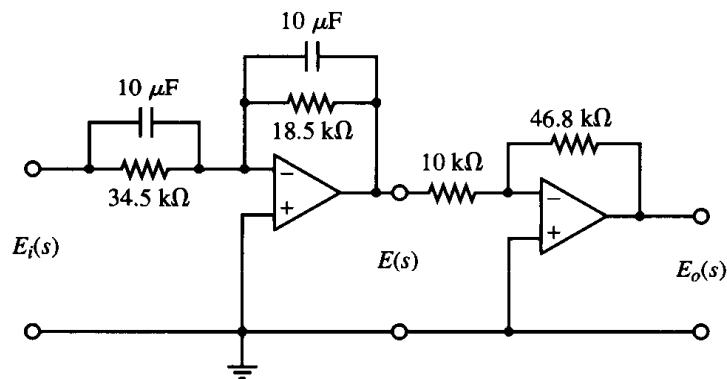


Figure 7-11
Lead compensator.

The static velocity error constant K_v is obtained from the expression

$$\begin{aligned} K_v &= \lim_{s \rightarrow 0} s G_c(s) G(s) \\ &= \lim_{s \rightarrow 0} \frac{s 18.7(s + 2.9)}{s(s + 2)(s + 5.4)} \\ &= 5.02 \text{ sec}^{-1} \end{aligned}$$

Note that the third closed-loop pole of the designed system is found by dividing the characteristic equation by the known factors as follows:

$$s(s + 2)(s + 5.4) + 18.7(s + 2.9) = (s + 2 + j2\sqrt{3})(s + 2 - j2\sqrt{3})(s + 3.4)$$

The foregoing compensation method enables us to place the dominant closed-loop poles at the desired points in the complex plane. The third pole at $s = 3.4$ is close to the added zero at $s = -2.9$. Therefore, the effect of this pole on the transient response is relatively small. Since no restriction has been imposed on the nondominant pole and no specification has been given concerning the value of the static velocity error coefficient, we conclude that the present design is satisfactory.

Comments. We may place the zero of the compensator at $s = -2$ and pole at $s = -4$ so that the angle contribution of the lead compensator is 30° . (In this case the zero of the lead compensator will cancel a pole of the plant, resulting in the second-order system, rather than the third-order system as we designed.) It can be seen that the K_v value in this case is 4 sec^{-1} . Other combinations can be selected that will yield 30° phase lead. (For different combinations of a zero and pole of the compensator that contribute 30° , the value of α will be different and the value of K_v will also be different.) Although a certain change in the value of K_v can be made by altering the pole-zero location of the lead compensator, if a large increase in the value of K_v is desired, then we must alter the lead compensator to a lag-lead compensator. (See Section 7-5 for lag-lead compensation.)

Comparison of step responses of the compensated and uncompensated systems. In what follows we shall examine the unit-step responses of the compensated and uncompensated systems with MATLAB.

The closed-loop transfer function of the compensated system is

$$\begin{aligned} \frac{C(s)}{R(s)} &= \frac{18.7(s + 2.9)}{s(s + 2)(s + 5.4) + 18.7(s + 2.9)} \\ &= \frac{18.7s + 54.23}{s^3 + 7.4s^2 + 29.5s + 54.23} \end{aligned}$$

Hence,

$$\begin{aligned} \text{numc} &= [0 \quad 0 \quad 18.7 \quad 54.23] \\ \text{denc} &= [1 \quad 7.4 \quad 29.5 \quad 54.23] \end{aligned}$$

For the uncompensated system, the closed-loop transfer function is

$$\frac{C(s)}{R(s)} = \frac{4}{s^2 + 2s + 4}$$

Hence,

```
num = [0 0 4]
den = [1 2 4]
```

MATLAB Program 7-1 produces the unit-step response curves for the two systems. The resulting plot is shown in Figure 7-12.

MATLAB Program 7-1

```
% ----- Unit-step response -----

% ***** Unit-step responses of compensated and uncompensated
% systems *****

numc = [0 0 18.7 54.23];
denc = [1 7.4 29.5 54.23];
num = [0 0 4];
den = [1 2 4];
t = 0:0.05:5;
[c1,x1,t] = step(numc,denc,t);
[c2,x2,t] = step(num,den,t);
plot(t,c1,t,c1,'o',t,c2,t,c2,'x')
grid
title('Unit-Step Responses of Compensated and Uncompensated Systems')
xlabel('t Sec')
ylabel('Outputs c1 and c2')
text(0.6,1.32,'Compensated system')
text(1.3,0.68,'Uncompensated system')
```

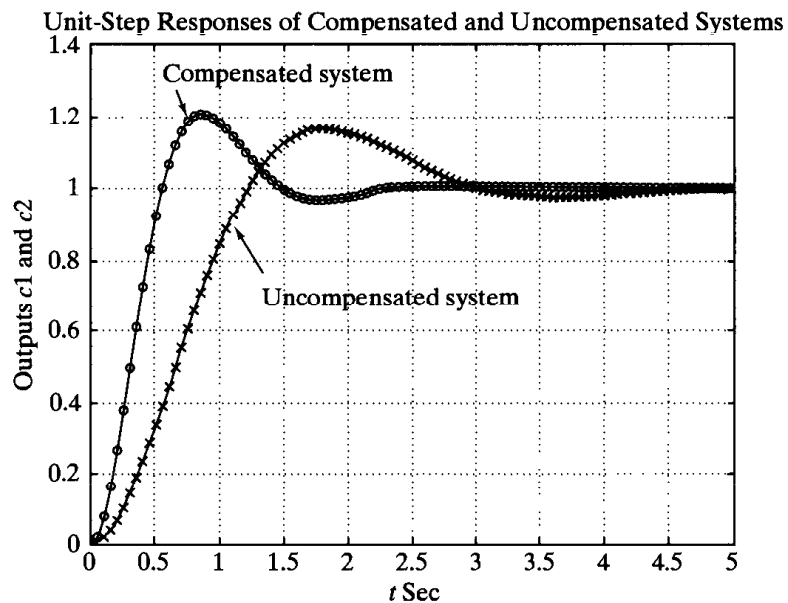


Figure 7-12
Unit-step responses
of compensated
and uncompen-
sated systems.

7-4 LAG COMPENSATION

Electronic lag compensator using operational amplifiers. The configuration of the electronic lag compensator using operational amplifiers is the same as that for the lead compensator shown in Figure 7-4. If we choose $R_2C_2 > R_1C_1$ in the circuit shown in Figure 7-4, it becomes a lag compensator. Referring to Figure 7-4, the transfer function of the lag compensator is given by

$$\frac{E_o(s)}{E_i(s)} = \hat{K}_c \beta \frac{Ts + 1}{\beta Ts + 1} = \hat{K}_c \frac{s + \frac{1}{T}}{s + \frac{1}{\beta T}}$$

where

$$T = R_1C_1, \quad \beta T = R_2C_2, \quad \beta = \frac{R_2C_2}{R_1C_1} > 1, \quad \hat{K}_c = \frac{R_4C_1}{R_3C_2}$$

Note that we use β instead of α in the above expressions. [In the lead compensator we used α to indicate the ratio $R_2C_2/(R_1C_1)$, which was less than 1, or $0 < \alpha < 1$.] In this chapter we always assume that $0 < \alpha < 1$ and $\beta > 1$.

Lag compensation techniques based on the root-locus approach. Consider the problem of finding a suitable compensation network for the case where the system exhibits satisfactory transient-response characteristics but unsatisfactory steady-state characteristics. Compensation in this case essentially consists of increasing the open-loop gain without appreciably changing the transient-response characteristics. This means that the root locus in the neighborhood of the dominant closed-loop poles should not be changed appreciably, but the open-loop gain should be increased as much as needed. This can be accomplished if a lag compensator is put in cascade with the given feedforward transfer function.

To avoid an appreciable change in the root loci, the angle contribution of the lag network should be limited to a small amount, say 5° . To assure this, we place the pole and zero of the lag network relatively close together and near the origin of the s plane. Then the closed-loop poles of the compensated system will be shifted only slightly from their original locations. Hence, the transient-response characteristics will be changed only slightly.

Consider a lag compensator $G_c(s)$, where

$$G_c(s) = \hat{K}_c \beta \frac{Ts + 1}{\beta Ts + 1} = \hat{K}_c \frac{s + \frac{1}{T}}{s + \frac{1}{\beta T}} \quad (7-2)$$

If we place the zero and pole of the lag compensator very close to each other, then at $s = s_1$, where s_1 is one of the dominant closed-loop poles, the magnitudes $s_1 + (1/T)$ and $s_1 + [1/(\beta T)]$ are almost equal, or

$$|G_c(s_1)| = \left| \hat{K}_c \frac{s_1 + \frac{1}{T}}{s_1 + \frac{1}{\beta T}} \right| \doteq \hat{K}_c$$

This implies that if gain $\hat{K}_c$ of the lag compensator is set equal to 1 then the transient-response characteristics will not be altered. (This means that the overall gain of the open-loop transfer function can be increased by a factor of β where $\beta > 1$.) If the pole and zero are placed very close to the origin, then the value of β can be made large. (A large value of β may be used, provided physical realization of the lag compensator is possible.) It is noted that the value of T must be large, but its exact value is not critical. However, it should not be too large in order to avoid difficulties in realizing the phase lag compensator by physical components.

An increase in the gain means an increase in the static error constants. If the open-loop transfer function of the uncompensated system is $G(s)$, then the static velocity error constant K_v of the uncompensated system is

$$K_v = \lim_{s \rightarrow 0} sG(s)$$

If the compensator is chosen as given by Equation (7-2), then for the compensated system with the open-loop transfer function $G_c(s)G(s)$ the static velocity error constant $\hat{K}_v$ becomes

$$\begin{aligned} \hat{K}_v &= \lim_{s \rightarrow 0} sG_c(s)G(s) \\ &= \lim_{s \rightarrow 0} G_c(s)K_v \\ &= \hat{K}_c \beta K_v \end{aligned}$$

Thus if the compensator is given by Equation (7-2), then the static velocity error constant is increased by a factor of $\hat{K}_c \beta$, where $\hat{K}_c$ is approximately unity.

Design procedures for lag compensation by the root-locus method. The procedure for designing lag compensators for the system shown in Figure 7-13 by the root-locus method may be stated as follows (we assume that the uncompensated system meets the transient-response specifications by simple gain adjustment; if this is not the case, refer to Section 7-5):

1. Draw the root-locus plot for the uncompensated system whose open-loop transfer function is $G(s)$. Based on the transient-response specifications, locate the dominant closed-loop poles on the root locus.

2. Assume the transfer function of the lag compensator to be

$$G_c(s) = \hat{K}_c \beta \frac{Ts + 1}{\beta Ts + 1} = \hat{K}_c \frac{s + \frac{1}{T}}{s + \frac{1}{\beta T}}$$

Then the open-loop transfer function of the compensated system becomes $G_c(s)G(s)$.

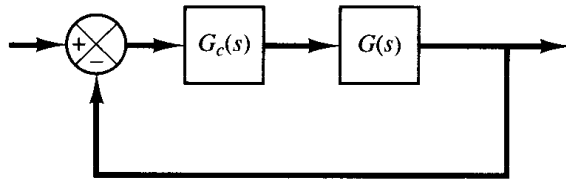


Figure 7-13
Control system.

3. Evaluate the particular static error constant specified in the problem.
4. Determine the amount of increase in the static error constant necessary to satisfy the specifications.
5. Determine the pole and zero of the lag compensator that produce the necessary increase in the particular static error constant without appreciably altering the original root loci. (Note that the ratio of the value of gain required in the specifications and the gain found in the uncompensated system is the required ratio between the distance of the zero from the origin and that of the pole from the origin.)
6. Draw a new root-locus plot for the compensated system. Locate the desired dominant closed-loop poles on the root locus. (If the angle contribution of the lag network is very small, that is, a few degrees, then the original and new root loci are almost identical. Otherwise, there will be a slight discrepancy between them. Then locate, on the new root locus, the desired dominant closed-loop poles based on the transient-response specifications.)
7. Adjust gain $\hat{K}_c$ of the compensator from the magnitude condition so that the dominant closed-loop poles lie at the desired location.

EXAMPLE 7-2

Consider the system shown in Figure 7-14(a). The feedforward transfer function is

$$G(s) = \frac{1.06}{s(s+1)(s+2)}$$

The root-locus plot for the system is shown in Figure 7-14(b). The closed-loop transfer function becomes

$$\begin{aligned} \frac{C(s)}{R(s)} &= \frac{1.06}{s(s+1)(s+2) + 1.06} \\ &= \frac{1.06}{(s+0.3307-j0.5864)(s+0.3307+j0.5864)(s+2.3386)} \end{aligned}$$

The dominant closed-loop poles are

$$s = -0.3307 \pm j0.5864$$

The damping ratio of the dominant closed-loop poles is $\zeta = 0.491$. The undamped natural frequency of the dominant closed-loop poles is 0.673 rad/sec. The static velocity error constant is 0.53 sec^{-1} .

It is desired to increase the static velocity error constant K_v to about 5 sec^{-1} without appreciably changing the location of the dominant closed-loop poles.

To meet this specification, let us insert a lag compensator as given by Equation (7-2) in cascade with the given feedforward transfer function. To increase the static velocity error constant

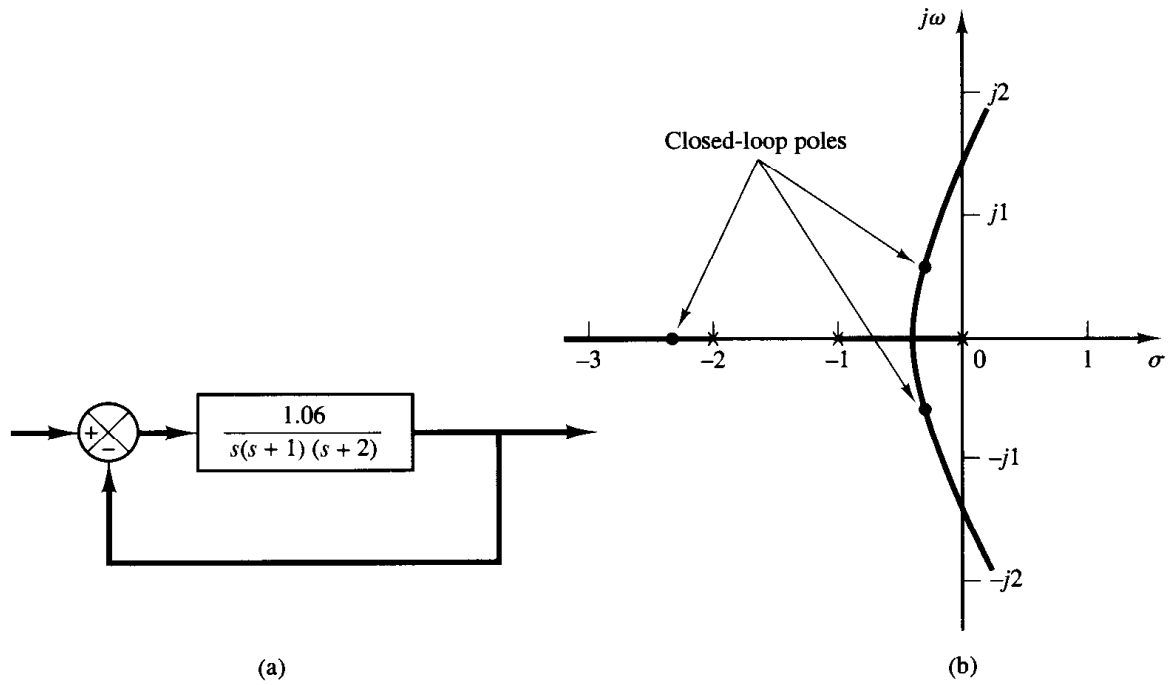


Figure 7-14
(a) Control system;
(b) root-locus plot.

by a factor of about 10, let us choose $\beta = 10$ and place the zero and pole of the lag compensator at $s = -0.05$ and $s = -0.005$, respectively. The transfer function of the lag compensator becomes

$$G_c(s) = \hat{K}_c \frac{s + 0.05}{s + 0.005}$$

The angle contribution of this lag network near a dominant closed-loop pole is about 4° . Because this angle contribution is not very small, there is a small change in the new root locus near the desired dominant closed-loop poles.

The open-loop transfer function of the compensated system then becomes

$$\begin{aligned} G_c(s)G(s) &= \hat{K}_c \frac{s + 0.05}{s + 0.005} \frac{1.06}{s(s + 1)(s + 2)} \\ &= \frac{K(s + 0.05)}{s(s + 0.005)(s + 1)(s + 2)} \end{aligned}$$

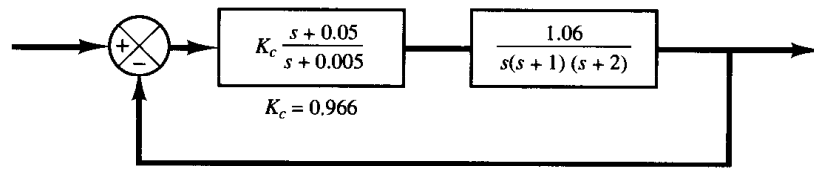
where

$$K = 1.06\hat{K}_c$$

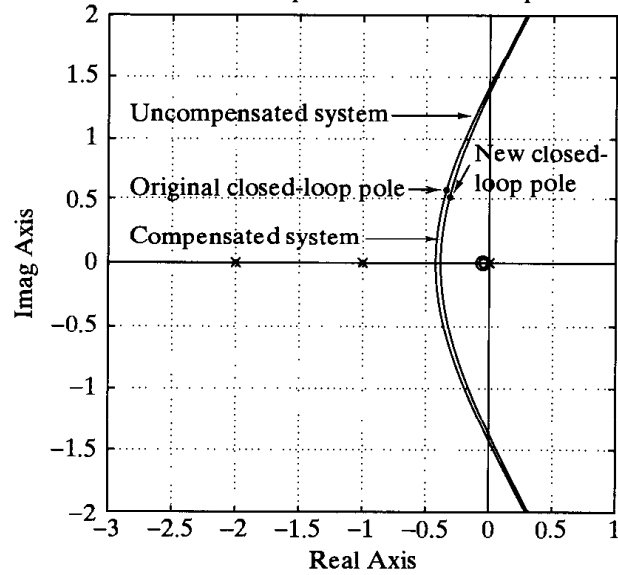
The block diagram of the compensated system is shown in Figure 7-15(a). The root-locus plot for the compensated system near the dominant closed-loop poles is shown in Figure 7-15(b), together with the original root-locus plot. Figure 7-15(c) shows the root-locus plot of the compensated system near the origin. The MATLAB program to generate the root-locus plots shown in Figures 7-15(b) and (c) is given in MATLAB Program 7-2.

If the damping ratio of the new dominant closed-loop poles is kept the same, then the poles are obtained from the new root-locus plot as follows:

$$s_1 = -0.31 + j0.55, \quad s_2 = -0.31 - j0.55$$



Root-Locus Plots of Compensated and Uncompensated Systems



Root-Locus Plot of Compensated System near the Origin

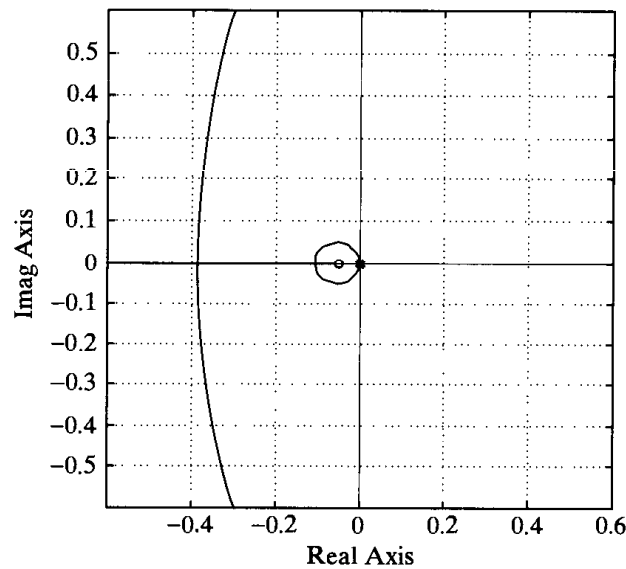


Figure 7-15

(a) Compensated system; (b) root-locus plots of the compensated system and the uncompensated system; (c) root-locus plot of compensated system near the origin.

MATLAB Program 7-2

```
% ----- Root Locus Plot -----

% ***** Root locus plots of the compensated system and
% uncompensated system *****

% ***** Enter the numerators and denominators of the
% compensated and uncompensated systems *****

numc = [0 0 0 1 0.05];
denc = [1 3.005 2.015 0.01 0];
num = [0 0 0 1.06];
den = [1 3 2 0];

% ***** Enter rlocus command. Plot the root loci of both
% systems *****

rlocus(numc,denc)
hold
Current plot held
rlocus(num,den)
v = [-3 1 -2 2]; axis(v); axis('square')
grid
text(-2,8,0.2,'Compensated system')
text(-2.8,1.2,'Uncompensated system')
text(-2.8,0.58,'Original closed-loop pole')
text(-0.1,0.85,'New closed-')
text(-0.1,0.62,'loop pole')
title('Root-Locus Plots of Compensated and Uncompensated Systems')

hold
Current plot released

% ***** Plot root loci of the compensated system near the origin *****

rlocus(numc,denc)
v = [-0.6 0.6 -0.6 0.6]; axis(v); axis('square')
grid
title('Root-Locus Plot of Compensated System near the Origin')
```

The open-loop gain K is

$$K = \left| \frac{s(s + 0.005)(s + 1)(s + 2)}{s + 0.05} \right|_{s = -0.31 + j0.55}$$

$$= 1.0235$$

Then the lag compensator gain $\hat{K}_c$ is determined as

$$\hat{K}_c = \frac{K}{1.06} = \frac{1.0235}{1.06} = 0.9656$$

Thus the transfer function of the lag compensator designed is

$$G_c(s) = 0.9656 \frac{s + 0.05}{s + 0.005} = 9.656 \frac{20s + 1}{200s + 1}$$

Then the compensated system has the following open-loop transfer function:

$$\begin{aligned} G_1(s) &= \frac{1.0235(s + 0.05)}{s(s + 0.005)(s + 1)(s + 2)} \\ &= \frac{5.12(20s + 1)}{s(200s + 1)(s + 1)(0.5s + 1)} \end{aligned}$$

The static velocity error constant K_v is

$$K_v = \lim_{s \rightarrow 0} sG_1(s) = 5.12 \text{ sec}^{-1}$$

In the compensated system, the static velocity error constant has increased to 5.12 sec^{-1} , or $5.12/0.53 = 9.66$ times the original value. (The steady-state error with ramp inputs has decreased to about 10% of that of the original system.) We have essentially accomplished the design objective of increasing the static velocity error constant to about 5 sec^{-1} .

Note that, since the pole and zero of the lag compensator are placed close together and are located very near the origin, their effect on the shape of the original root loci has been small. Except for the presence of a small closed root locus near the origin, the root loci of the compensated and the uncompensated systems are very similar to each other. However, the value of the static velocity error constant of the compensated system is 9.66 times greater than that of the uncompensated system.

The two other closed-loop poles for the compensated system are found as follows:

$$s_3 = -2.326, \quad s_4 = -0.0549$$

The addition of the lag compensator increases the order of the system from 3 to 4, adding one additional closed-loop pole close to the zero of the lag compensator. (The added closed-loop pole at $s = -0.0549$ is close to the zero at $s = -0.05$.) Such a pair of a zero and pole creates a long tail of small amplitude in the transient response, as we will see later in the unit-step response. Since the pole at $s = -2.326$ is very far from the $j\omega$ axis compared with the dominant closed-loop poles, the effect of this pole on the transient response is also small. Therefore, we may consider the closed-loop poles at $s = -0.31 \pm j0.55$ to be the dominant closed-loop poles.

The undamped natural frequency of the dominant closed-loop poles of the compensated system is 0.631 rad/sec . This value is about 6% less than the original value, 0.673 rad/sec . This implies that the transient response of the compensated system is slower than that of the original system. The response will take a longer time to settle down. The maximum overshoot in the step response will increase in the compensated system. If such adverse effects can be tolerated, the lag compensation as discussed here presents a satisfactory solution to the given design problem.

Next, we shall compare the unit-ramp responses of the compensated system against the uncompensated system and verify that the steady-state performance is much better in the compensated system than the uncompensated system.

To obtain the unit-ramp response with MATLAB, we use the step command for the system $C(s)/[sR(s)]$. Since $C(s)/[sR(s)]$ for the compensated system is

$$\begin{aligned} \frac{C(s)}{sR(s)} &= \frac{1.0235(s + 0.05)}{s[s(s + 0.005)(s + 1)(s + 2) + 1.0235(s + 0.05)]} \\ &= \frac{1.0235s + 0.0512}{s^5 + 3.005s^4 + 2.015s^3 + 1.0335s^2 + 0.0512s} \end{aligned}$$

we have

$$\begin{aligned} \text{numc} &= [0 \quad 0 \quad 0 \quad 0 \quad 1.0235 \quad 0.0512] \\ \text{denc} &= [1 \quad 3.005 \quad 2.015 \quad 1.0335 \quad 0.0512 \quad 0] \end{aligned}$$

Also, $C(s)/[sR(s)]$ for the uncompensated system is

$$\begin{aligned}\frac{C(s)}{sR(s)} &= \frac{1.06}{s[s(s+1)(s+2) + 1.06]} \\ &= \frac{1.06}{s^4 + 3s^3 + 2s^2 + 1.06s}\end{aligned}$$

Hence,

$$\begin{aligned}\text{num} &= [0 \quad 0 \quad 0 \quad 0 \quad 1.06] \\ \text{den} &= [1 \quad 3 \quad 2 \quad 1.06 \quad 0]\end{aligned}$$

MATLAB Program 7-3 produces the plot of the unit-ramp response curves. Figure 7-16 shows the result. Clearly, the compensated system shows much smaller steady-state error (one-tenth of the original steady-state error) in following the unit-ramp input.

MATLAB Program 7-4 gives the unit-step response curves of the compensated and uncompensated systems. The unit-step response curves are shown in Figure 7-17. Notice that the

MATLAB Program 7-3

```
% ----- Unit ramp response -----

% ***** Unit-ramp responses of compensated system and
% uncompensated system *****

% ***** Unit-ramp response will be obtained as the unit-step
% response of C(s)/[sR(s)] *****

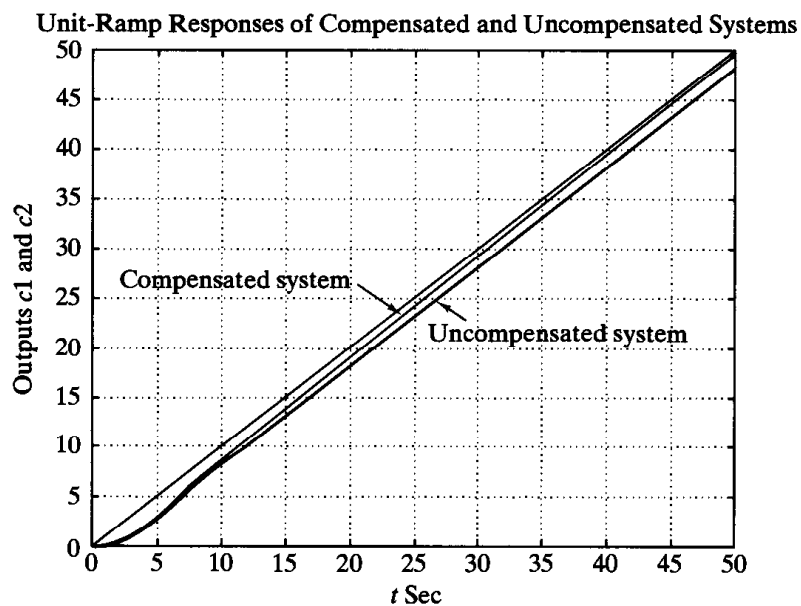
% ***** Enter the numerators and denominators of C1(s)/[sR(s)]
% and C2(s)/[sR(s)], where C1(s) and C2(s) are Laplace
% transforms of the outputs of the compensated and un-
% compensated systems, respectively. *****

numc = [0 0 0 0 1.0235 0.0512];
denc = [1 3.005 2.015 1.0335 0.0512 0];
num = [0 0 0 0 1.06];
den = [1 3 2 1.06 0];

% ***** Specify the time range (such as t = 0:0.1:50) and enter
% step command and plot command. *****

t = 0:0.1:50;
[c1,x1,t] = step(numc,denc,t);
[c2,x2,t] = step(num,den,t);
plot(t,c1,'-',t,c2,'.',t,t,'- -')
grid
text(2.2,27,'Compensated system');
text(26,21.3,'Uncompensated system')
title('Unit-Ramp Responses of Compensated and Uncompensated Systems')
xlabel('t Sec');
ylabel('Outputs c1 and c2')
```

Figure 7-16
Unit-ramp responses
of compensated
and uncompen-
sated systems.



MATLAB Program 7-4

```
% ----- Unit-step response -----

% ***** Unit-step responses of compensated system and
% uncompensated system *****

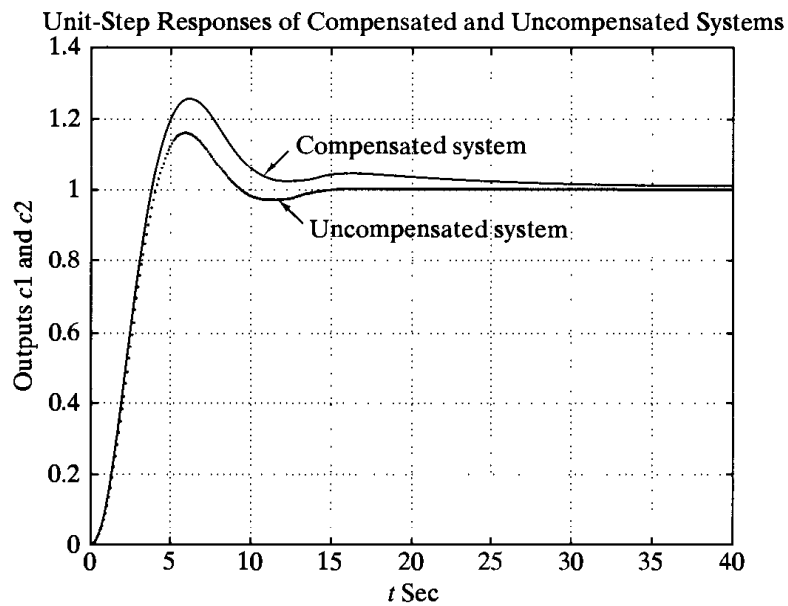
% ***** Enter the numerators and denominators of the
% compensated and uncompensated systems *****

numc = [0 0 0 1.0235 0.0512];
denc = [1 3.005 2.015 1.0335 0.0512];
num = [0 0 0 1.06];
den = [1 3 2 1.06];

% ***** Specify the time range (such as t = 0:0.1:40) and enter
% step command and plot command. *****

t = 0:0.1:40;
[c1,x1,t] = step(numc,denc,t);
[c2,x2,t] = step(num,den,t);
plot(t,c1,'-',t,c2,'.')
grid
text(13,1.12,'Compensated system')
text(13.6,0.88,'Uncompensated system')
title('Unit-Step Responses of Compensated and Uncompensated Systems')
xlabel('t Sec')
ylabel('Outputs c1 and c2')
```

Figure 7-17
Unit-step responses
of compensated
and uncompen-
sated systems.



lag-compensated system exhibits a larger maximum overshoot and slower response than the original uncompensated system. Notice that a pair of the pole at $s = -0.0549$ and zero at $s = -0.05$ generates a long tail of small amplitude in the transient response. If a larger maximum overshoot and a slower response are not desired, we need to use a lag-lead compensator as presented in Section 7-5.

7-5 LAG-LEAD COMPENSATION

Lead compensation basically speeds up the response and increases the stability of the system. Lag compensation improves the steady-state accuracy of the system, but reduces the speed of the response.

If improvements in both transient response and steady-state response are desired, then both a lead compensator and a lag compensator may be used simultaneously. Rather than introducing both a lead compensator and a lag compensator as separate elements, however, it is economical to use a single lag-lead compensator.

Lag-lead compensation combines the advantages of lag and lead compensations. Since the lag-lead compensator possesses two poles and two zeros, such a compensation increases the order of the system by 2, unless cancellation of pole(s) and zero(s) occurs in the compensated system.

Electronic lag-lead compensator using operational amplifiers. Figure 7-18 shows an electronic lag-lead compensator using operational amplifiers. The transfer function for this compensator may be obtained as follows: The complex impedance Z_1 is given by

$$\frac{1}{Z_1} = \frac{1}{R_1 + \frac{1}{C_1 s}} + \frac{1}{R_3}$$

or

$$Z_1 = \frac{(R_1 C_1 s + 1) R_3}{(R_1 + R_3) C_1 s + 1}$$

Similarly, complex impedance Z_2 is given by

$$Z_2 = \frac{(R_2 C_2 s + 1) R_4}{(R_2 + R_4) C_2 s + 1}$$

Hence, we have

$$\frac{E(s)}{E_i(s)} = -\frac{Z_2}{Z_1} = -\frac{R_4}{R_3} \frac{(R_1 + R_3) C_1 s + 1}{R_1 C_1 s + 1} \cdot \frac{R_2 C_2 s + 1}{(R_2 + R_4) C_2 s + 1}$$

The sign inverter has the transfer function

$$\frac{E_o(s)}{E(s)} = -\frac{R_6}{R_5}$$

Thus the transfer function of the compensator shown in Figure 7-18 is

$$\frac{E_o(s)}{E_i(s)} = \frac{E_o(s)}{E(s)} \frac{E(s)}{E_i(s)} = \frac{R_4 R_6}{R_3 R_5} \left[\frac{(R_1 + R_3) C_1 s + 1}{R_1 C_1 s + 1} \right] \left[\frac{R_2 C_2 s + 1}{(R_2 + R_4) C_2 s + 1} \right] \quad (7-3)$$

Let us define

$$T_1 = (R_1 + R_3) C_1, \quad \frac{T_1}{\gamma} = R_1 C_1, \quad T_2 = R_2 C_2, \quad \beta T_2 = (R_2 + R_4) C_2$$

Then Equation (7-3) becomes

$$\frac{E_o(s)}{E_i(s)} = K_c \frac{\beta}{\gamma} \left(\frac{T_1 s + 1}{\frac{T_1}{\gamma} s + 1} \right) \left(\frac{T_2 s + 1}{\beta T_2 s + 1} \right) = K_c \frac{\left(s + \frac{1}{T_1} \right) \left(s + \frac{1}{T_2} \right)}{\left(s + \frac{\gamma}{T_1} \right) \left(s + \frac{1}{\beta T_2} \right)} \quad (7-4)$$

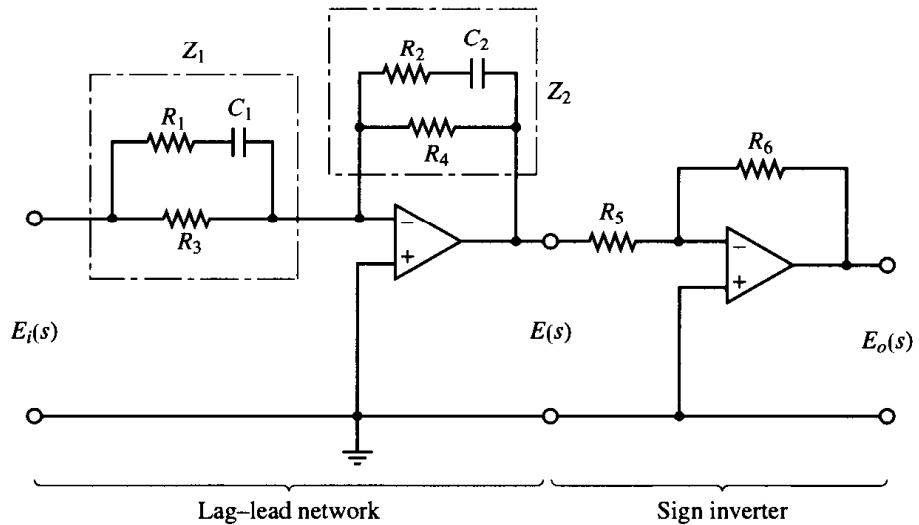


Figure 7-18
Lag-lead
compensator.

where

$$\gamma = \frac{R_1 + R_3}{R_1} > 1, \quad \beta = \frac{R_2 + R_4}{R_2} > 1, \quad K_c = \frac{R_2 R_4 R_6}{R_1 R_3 R_5} \frac{R_1 + R_3}{R_2 + R_4}$$

Note that β is often chosen to be equal to γ .

Lag-lead compensation techniques based on the root-locus approach. Consider the system shown in Figure 7-19. Assume that we use the lag-lead compensator:

$$G_c(s) = K_c \frac{\beta}{\gamma} \frac{(T_1 s + 1)(T_2 s + 1)}{\left(\frac{T_1}{\gamma} s + 1\right)(\beta T_2 s + 1)} = K_c \left(\frac{s + \frac{1}{T_1}}{s + \frac{\gamma}{T_1}} \right) \left(\frac{s + \frac{1}{T_2}}{s + \frac{1}{\beta T_2}} \right) \quad (7-5)$$

where $\beta > 1$ and $\gamma > 1$. (Consider K_c to belong to the lead portion of the lag-lead compensator.)

In designing lag-lead compensators, we consider two cases where $\gamma \neq \beta$ and $\gamma = \beta$.

Case 1. $\gamma \neq \beta$. In this case, the design process is a combination of the design of the lead compensator and that of the lag compensator. The design procedure for the lag-lead compensator follows:

1. From the given performance specifications, determine the desired location for the dominant closed-loop poles.
2. Using the uncompensated open-loop transfer function $G(s)$, determine the angle deficiency ϕ if the dominant closed-loop poles are to be at the desired location. The phase-lead portion of the lag-lead compensator must contribute this angle ϕ .
3. Assuming that we later choose T_2 sufficiently large so that the magnitude of the lag portion

$$\left| \frac{s_1 + \frac{1}{T_2}}{s_1 + \frac{1}{\beta T_2}} \right|$$

is approximately unity, where $s = s_1$ is one of the dominant closed-loop poles, choose the values of T_1 and γ from the requirement that

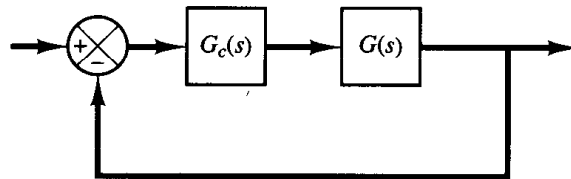


Figure 7-19
Control system.

$$\left/ \begin{array}{c} s_1 + \frac{1}{T_1} \\ s_1 + \frac{\gamma}{T_1} \end{array} \right/ = \phi$$

The choice of T_1 and γ is not unique. (Infinitely many sets of T_1 and γ are possible.) Then determine the value of K_c from the magnitude condition:

$$\left| K_c \frac{s_1 + \frac{1}{T_1}}{s_1 + \frac{\gamma}{T_1}} G(s_1) \right| = 1$$

4. If the static velocity error constant K_v is specified, determine the value of β to satisfy the requirement for K_v . The static velocity error constant K_v is given by

$$\begin{aligned} K_v &= \lim_{s \rightarrow 0} s G_c(s) G(s) \\ &= \lim_{s \rightarrow 0} s K_c \left(\frac{s + \frac{1}{T_1}}{s + \frac{\gamma}{T_1}} \right) \left(\frac{s + \frac{1}{T_2}}{s + \frac{1}{\beta T_2}} \right) G(s) \\ &= \lim_{s \rightarrow 0} s K_c \frac{\beta}{\gamma} G(s) \end{aligned}$$

where K_c and γ are already determined in step 3. Hence, given the value of K_v , the value of β can be determined from this last equation. Then, using the value of β thus determined, choose the value of T_2 such that

$$\begin{aligned} &\left| \frac{s_1 + \frac{1}{T_2}}{s_1 + \frac{1}{\beta T_2}} \right| \doteq 1 \\ &-5^\circ < \left/ \begin{array}{c} s_1 + \frac{1}{T_2} \\ s_1 + \frac{1}{\beta T_2} \end{array} \right/ < 0^\circ \end{aligned}$$

(The preceding design procedure is illustrated in Example 7-3.)

Case 2. $\gamma = \beta$. If $\gamma = \beta$ is required in Equation (7-5), then the preceding design procedure for the lag-lead compensator may be modified as follows:

1. From the given performance specifications, determine the desired location for the dominant closed-loop poles.

2. The lag-lead compensator given by Equation (7-5) is modified to

$$G_c(s) = K_c \frac{(T_1s + 1)(T_2s + 1)}{\left(\frac{T_1}{\beta}s + 1\right)(\beta T_2s + 1)} = K_c \frac{\left(s + \frac{1}{T_1}\right)\left(s + \frac{1}{T_2}\right)}{\left(s + \frac{\beta}{T_1}\right)\left(s + \frac{1}{\beta T_2}\right)} \quad (7-6)$$

where $\beta > 1$. The open-loop transfer function of the compensated system is $G_c(s)G(s)$. If the static velocity error constant K_v is specified, determine the value of constant K_c from the following equation:

$$\begin{aligned} K_v &= \lim_{s \rightarrow 0} sG_c(s)G(s) \\ &= \lim_{s \rightarrow 0} sK_cG(s) \end{aligned}$$

3. To have the dominant closed-loop poles at the desired location, calculate the angle contribution ϕ needed from the phase lead portion of the lag-lead compensator.

4. For the lag-lead compensator, we later choose T_2 sufficiently large so that

$$\left| \frac{s_1 + \frac{1}{T_2}}{s_1 + \frac{1}{\beta T_2}} \right|$$

is approximately unity, where $s = s_1$ is one of the dominant closed-loop poles. Determine the values of T_1 and β from the magnitude and angle conditions:

$$\left| K_c \frac{s_1 + \frac{1}{T_1}}{s_1 + \frac{\beta}{T_1}} G(s_1) \right| = 1$$

$$\angle \frac{s_1 + \frac{1}{T_1}}{s_1 + \frac{\beta}{T_1}} = \phi$$

5. Using the value of β just determined, choose T_2 so that

$$\left| \frac{s_1 + \frac{1}{T_2}}{s_1 + \frac{1}{\beta T_2}} \right| \doteq 1$$

$$-5^\circ < \angle \frac{s_1 + \frac{1}{T_2}}{s_1 + \frac{1}{\beta T_2}} < 0^\circ$$

The value of βT_2 , the largest time constant of the lag-lead compensator, should not be too large to be physically realized. (An example of the design of the lag-lead compensator when $\gamma = \beta$ is given in Example 7-4).

EXAMPLE 7-3

Consider the control system shown in Figure 7-20. The feedforward transfer function is

$$G(s) = \frac{4}{s(s + 0.5)}$$

This system has closed-loop poles at

$$s = -0.2500 \pm j1.9843$$

The damping ratio is 0.125, the undamped natural frequency is 2 rad/sec, and the static velocity error constant is 8 sec^{-1} .

It is desired to make the damping ratio of the dominant closed-loop poles equal to 0.5 and to increase the undamped natural frequency to 5 rad/sec and the static velocity error constant to 80 sec^{-1} . Design an appropriate compensator to meet all the performance specifications.

Let us assume that we use a lag-lead compensator having the transfer function

$$G_c(s) = K_c \left(\frac{s + \frac{1}{T_1}}{s + \frac{\gamma}{T_1}} \right) \left(\frac{s + \frac{1}{T_2}}{s + \frac{1}{\beta T_2}} \right) \quad (\gamma > 1, \beta > 1)$$

where γ is not equal to β . Then the compensated system will have the transfer function

$$G_c(s)G(s) = K_c \left(\frac{s + \frac{1}{T_1}}{s + \frac{\gamma}{T_1}} \right) \left(\frac{s + \frac{1}{T_2}}{s + \frac{1}{\beta T_2}} \right) G(s)$$

From the performance specifications, the dominant closed-loop poles must be at

$$s = -2.50 \pm j4.33$$

Since

$$\left. \frac{4}{s(s + 0.5)} \right|_{s = -2.50 + j4.33} = -235^\circ$$

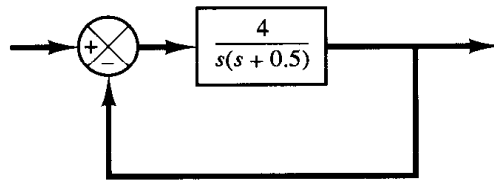


Figure 7-20
Control system.

the phase lead portion of the lag-lead compensator must contribute 55° so that the root locus passes through the desired location of the dominant closed-loop poles.

To design the phase lead portion of the compensator, we first determine the location of the zero and pole that will give 55° contribution. There are many possible choices, but we shall here choose the zero at $s = -0.5$ so that this zero will cancel the pole at $s = -0.5$ of the plant. Once the zero is chosen, the pole can be located such that the angle contribution is 55° . By simple calculation or graphical analysis, the pole must be located at $s = -5.021$. Thus, the phase lead portion of the lag-lead compensator becomes

$$K_c \frac{s + \frac{1}{T_1}}{s + \frac{\gamma}{T_1}} = K_c \frac{s + 0.5}{s + 5.021}$$

Thus

$$T_1 = 2, \quad \gamma = \frac{5.021}{0.5} = 10.04$$

Next we determine the value of K_c from the magnitude condition:

$$\left| K_c \frac{s + 0.5}{s + 5.021} \frac{4}{s(s + 0.5)} \right|_{s=-2.5+j4.33} = 1$$

Hence,

$$K_c = \left| \frac{(s + 5.021)s}{4} \right|_{s=-2.5+j4.33} = 6.26$$

The phase lag portion of the compensator can be designed as follows: First the value of β is determined to satisfy the requirement on the static velocity error constant:

$$\begin{aligned} K_v &= \lim_{s \rightarrow 0} s G_c(s) G(s) = \lim_{s \rightarrow 0} s K_c \frac{\beta}{\gamma} G(s) \\ &= \lim_{s \rightarrow 0} s (6.26) \frac{\beta}{10.04} \frac{4}{s(s + 0.5)} = 4.988\beta = 80 \end{aligned}$$

Hence, β is determined as

$$\beta = 16.04$$

Finally, we choose the value of T_2 large enough so that

$$\left| \frac{s + \frac{1}{T_2}}{s + \frac{1}{16.04T_2}} \right|_{s=-2.5+j4.33} \doteq 1$$

and

$$-5^\circ < \left| \frac{s + \frac{1}{T_2}}{s + \frac{1}{16.04T_2}} \right|_{s=-2.5+j4.33} < 0^\circ$$

Since $T_2 \doteq 5$ (or any number greater than 5) satisfies the above two requirements, we may choose

$$T_2 = 5$$

Now the transfer function of the designed lag-lead compensator is given by

$$\begin{aligned} G_c(s) &= (6.26) \left(\frac{s + \frac{1}{2}}{s + \frac{10.04}{2}} \right) \left(\frac{s + \frac{1}{5}}{s + \frac{1}{16.04 \times 5}} \right) \\ &= 6.26 \left(\frac{s + 0.5}{s + 5.02} \right) \left(\frac{s + 0.2}{s + 0.01247} \right) \\ &= \frac{10(2s + 1)(5s + 1)}{(0.1992s + 1)(80.19s + 1)} \end{aligned}$$

The compensated system will have the open-loop transfer function

$$G_c(s)G(s) = \frac{25.04(s + 0.2)}{s(s + 5.02)(s + 0.01247)}$$

Because of the cancellation of the $(s + 0.5)$ terms, the compensated system is a third-order system. (Mathematically, this cancellation is exact, but practically such cancellation will not be exact because some approximations are usually involved in deriving the mathematical model of the system and, as a result, the time constants are not precise.) The root-locus plot of the compensated system is shown in Figure 7-21(a). An enlarged view of the root-locus plot near the origin is shown in Figure 7-21(b). Because the angle contribution of the phase lag portion of the lag-lead compensator is quite small, there is only a small change in the location of the dominant closed-loop poles from the desired location, $s = -2.5 \pm j4.33$. In fact, the new closed-loop poles are located

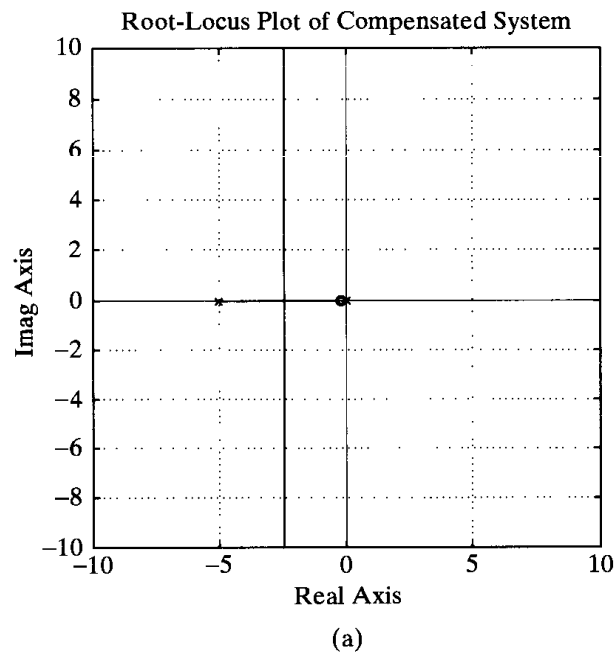


Figure 7-21
(a) Root-locus plot of the compensated system; (b) root-locus plot near the origin.

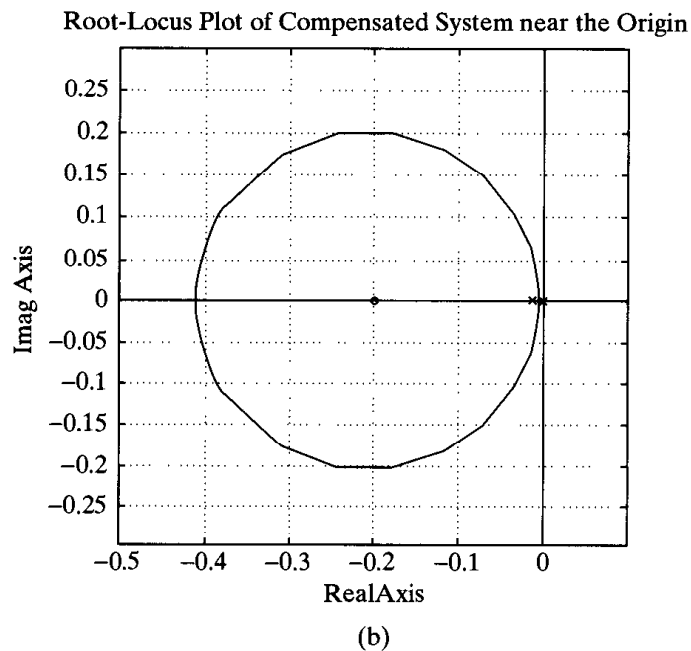


Figure 7-21
(Continued)

at $s = -2.4123 \pm j4.2756$. (The new damping ratio is $\zeta = 0.491$.) Therefore, the compensated system meets all the required performance specifications. The third closed-loop pole of the compensated system is located at $s = -0.2078$. Since this closed-loop pole is very close to the zero at $s = -0.2$, the effect of this pole on the response is small. (Note that, in general, if a pole and a zero lie close to each other on the negative real axis near the origin, then such a pole-zero combination will yield a long tail of small amplitude in the transient response.)

The unit-step response curves and unit-ramp response curves before and after compensation are shown in Figure 7-22.

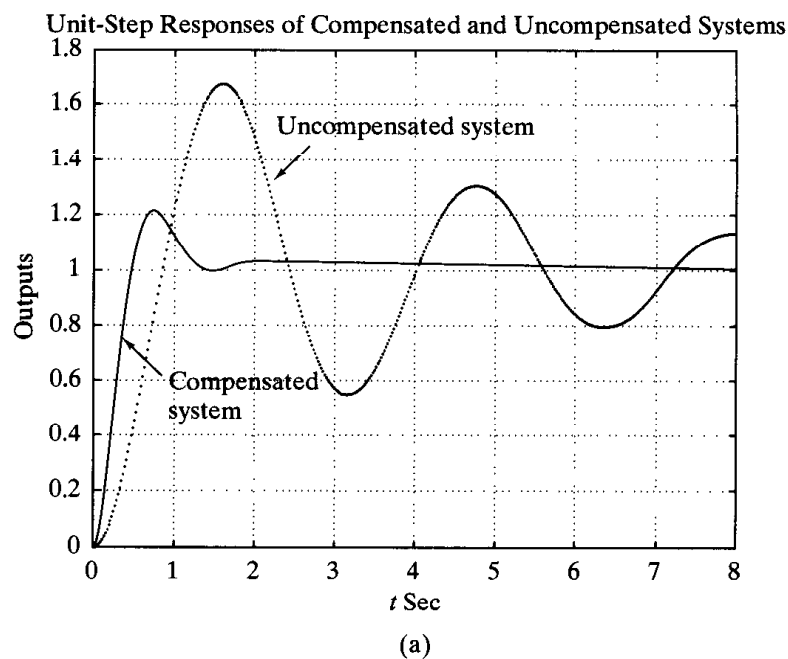


Figure 7-22
Transient response curves for the compensated system and the uncompensated system. (a) Unit-step response curves; (b) unit-ramp response curves.

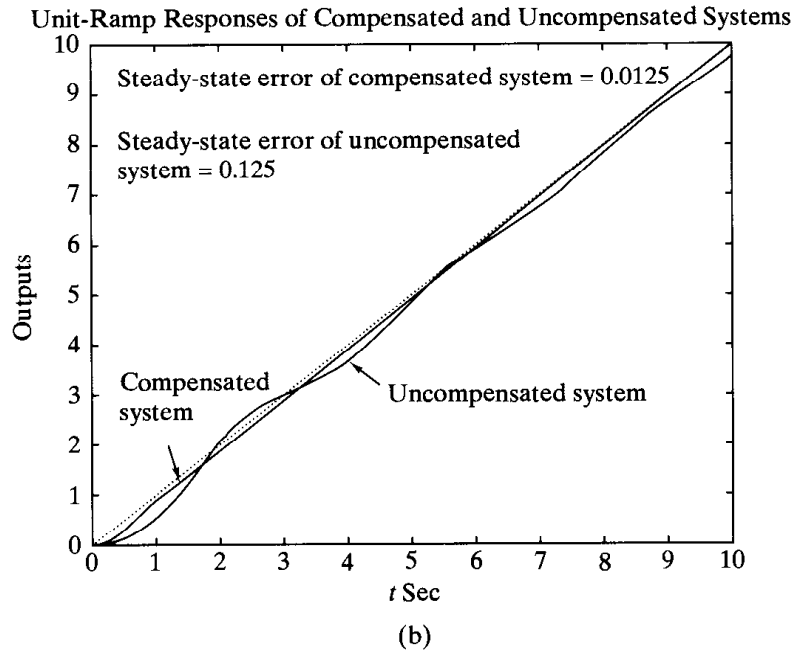


Figure 7-22
(Continued)

EXAMPLE 7-4

Consider the control system of Example 7-3. Suppose that we use a lag-lead compensator of the form given by Equation (7-6), or

$$G_c(s) = K_c \frac{\left(s + \frac{1}{T_1}\right)\left(s + \frac{1}{T_2}\right)}{\left(s + \frac{\beta}{T_1}\right)\left(s + \frac{1}{\beta T_2}\right)} \quad (\beta > 1)$$

Assuming the specifications are the same as those given in Example 7-3, design a compensator $G_c(s)$.

The desired locations for the dominant closed-loop poles are at

$$s = -2.50 \pm j4.33$$

The open-loop transfer function of the compensated system is

$$G_c(s)G(s) = K_c \frac{\left(s + \frac{1}{T_1}\right)\left(s + \frac{1}{T_2}\right)}{\left(s + \frac{\beta}{T_1}\right)\left(s + \frac{1}{\beta T_2}\right)} \cdot \frac{4}{s(s + 0.5)}$$

Since the requirement on the static velocity error constant K_v is 80 sec^{-1} , we have

$$K_v = \lim_{s \rightarrow 0} sG_c(s)G(s) = \lim_{s \rightarrow 0} K_c \frac{4}{0.5} = 8K_c = 80$$

Thus

$$K_c = 10$$

The time constant T_1 and the value of β are determined from

$$\left| \frac{s + \frac{1}{T_1}}{s + \frac{\beta}{T_1}} \right| \left| \frac{40}{s(s + 0.5)} \right|_{s = -2.5 + j4.33} = \left| \frac{s + \frac{1}{T_1}}{s + \frac{\beta}{T_1}} \right| \frac{8}{4.77} = 1$$

$$\left| \frac{s + \frac{1}{T_1}}{s + \frac{\beta}{T_1}} \right|_{s = -2.5 + j4.33} = 55^\circ$$

Referring to Figure 7-23, we can easily locate points A and B such that

$$\angle APB = 55^\circ, \quad \frac{\overline{PA}}{\overline{PB}} = \frac{4.77}{8}$$

(Use a graphical approach or a trigonometric approach.) The result is

$$\overline{AO} = 2.38, \quad \overline{BO} = 8.34$$

or

$$T_1 = \frac{1}{2.38} = 0.420, \quad \beta = 8.34T_1 = 3.503$$

The phase lead portion of the lag-lead network thus becomes

$$10 \left(\frac{s + 2.38}{s + 8.34} \right)$$

For the phase lag portion, we may choose

$$T_2 = 10$$

Then

$$\frac{1}{\beta T_2} = \frac{1}{3.503 \times 10} = 0.0285$$

Thus, the lag-lead compensator becomes

$$G_c(s) = (10) \left(\frac{s + 2.38}{s + 8.34} \right) \left(\frac{s + 0.1}{s + 0.0285} \right)$$

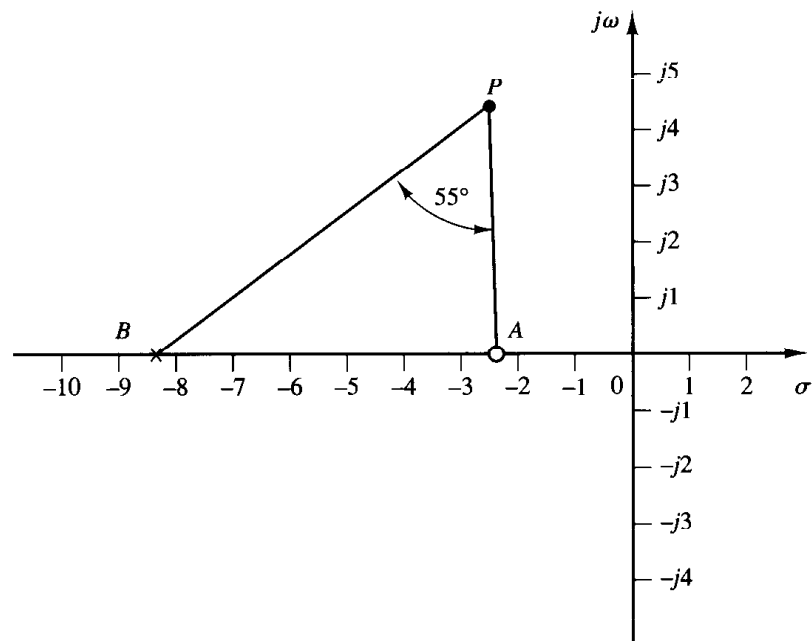


Figure 7-23
Determination of
the desired pole-
zero location.

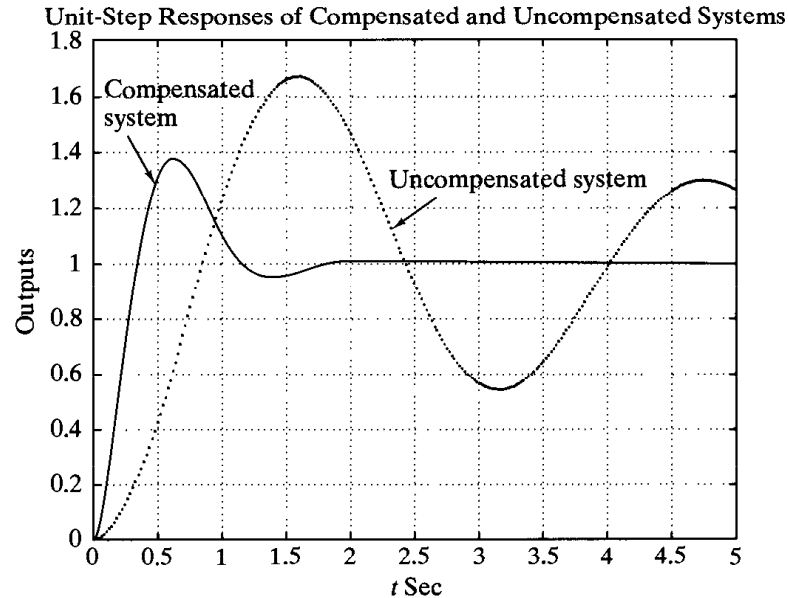
The compensated system will have the open-loop transfer function

$$G_c(s)G(s) = \frac{40(s + 2.38)(s + 0.1)}{(s + 8.34)(s + 0.0285)s(s + 0.5)}$$

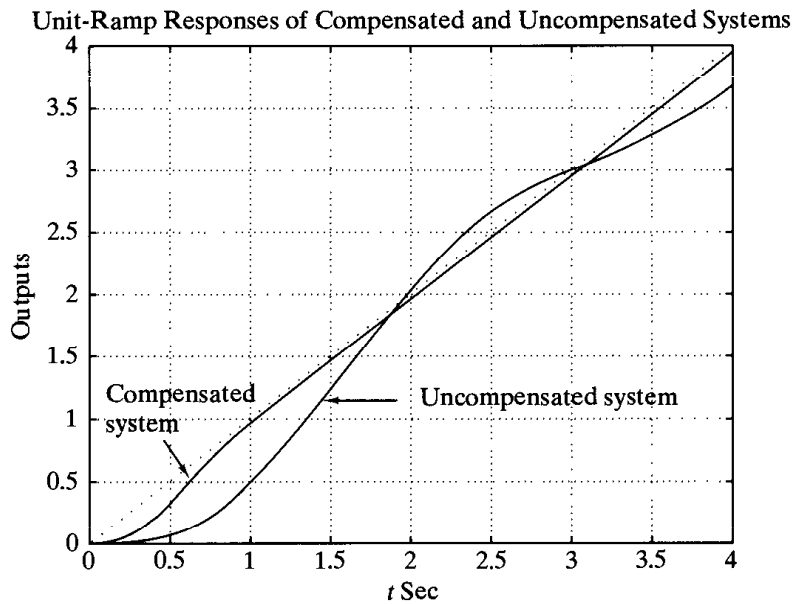
No cancellation occurs in this case, and the compensated system is of fourth order. Because the angle contribution of the phase lag portion of the lag-lead network is quite small, the dominant closed-loop poles are located very near the desired location. In fact, the dominant closed-loop poles are located at $s = -2.4539 \pm j4.3099$. The two other closed-loop poles are located at

$$s = -0.1003, \quad s = -3.8604$$

Since the closed-loop pole at $s = -0.1003$ is very close to a zero at $s = -0.1$, they almost cancel each other. Thus, the effect of this closed-loop pole is very small. The remaining closed-loop pole



(a)



(a)

Figure 7-24

(a) Unit-step response curves for the compensated and uncompensated systems; (b) unit-ramp response curves for both systems.

($s = -3.8604$) does not quite cancel the zero at $s = -2.4$. The effect of this zero is to cause a larger overshoot in the step response than a similar system without such a zero. The unit-step response curves of the compensated and uncompensated systems are shown in Figure 7-24(a). The unit-ramp response curves for both systems are depicted in Figure 7-24(b).

EXAMPLE PROBLEMS AND SOLUTIONS

- A-7-1.** Obtain the transfer function of the mechanical system shown in Figure 7-25. Assume that the displacement x_i is the input and displacement x_o is the output of the system.

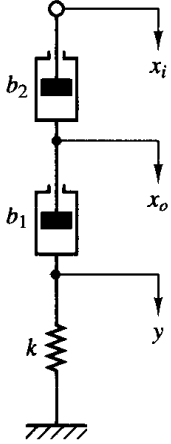


Figure 7-25
Mechanical system.

Solution. From the diagram we obtain the following equations of motion:

$$b_2(\dot{x}_i - \dot{x}_o) = b_1(\dot{x}_o - \dot{y})$$

$$b_1(\dot{x}_o - \dot{y}) = ky$$

Taking the Laplace transforms of these two equations, assuming zero initial conditions, and then eliminating $Y(s)$, we obtain

$$\frac{X_o(s)}{X_i(s)} = \frac{b_2}{b_1 + b_2} \frac{\frac{b_1}{k}s + 1}{\frac{b_2}{b_1 + b_2} \frac{b_1}{k}s + 1}$$

This is the transfer function between $X_o(s)$ and $X_i(s)$. By defining

$$\frac{b_1}{k} = T, \quad \frac{b_2}{b_1 + b_2} = \alpha < 1$$

we obtain

$$\frac{X_o(s)}{X_i(s)} = \alpha \frac{Ts + 1}{\alpha Ts + 1} = \frac{s + \frac{1}{T}}{s + \frac{1}{\alpha T}}$$

This mechanical system is a mechanical lead network.

- A-7-2.** Obtain the transfer function of the mechanical system shown in Figure 7-26. Assume that the displacement x_i is the input and displacement x_o is the output.

Solution. The equations of motion for this system are

$$b_2(\dot{x}_i - \dot{x}_o) + k_2(x_i - x_o) = b_1(\dot{x}_o - \dot{y})$$

$$b_1(\dot{x}_o - \dot{y}) = ky$$

By taking the Laplace transforms of these two equations, assuming zero initial conditions, we obtain

$$b_2[sX_i(s) - sX_o(s)] + k_2[X_i(s) - X_o(s)] = b_1[sX_o(s) - sY(s)]$$

$$b_1[sX_o(s) - sY(s)] = k_1Y(s)$$

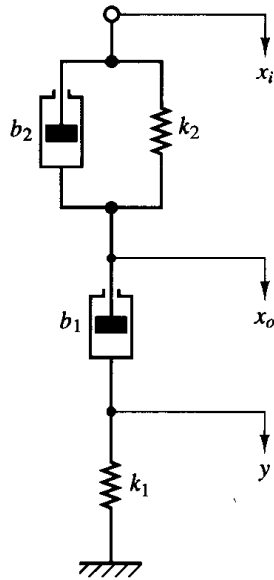


Figure 7-26
Mechanical system.

If we eliminate $Y(s)$ from the last two equations, the transfer function $X_o(s)/X_i(s)$ can be obtained as

$$\frac{X_o(s)}{X_i(s)} = \frac{\left(\frac{b_1}{k_1}s + 1\right)\left(\frac{b_2}{k_2}s + 1\right)}{\left(\frac{b_1}{k_1}s + 1\right)\left(\frac{b_2}{k_2}s + 1\right) + \frac{b_1}{k_2}s}$$

Define

$$T_1 = \frac{b_1}{k_1}, \quad T_2 = \frac{b_2}{k_2}, \quad \frac{b_1}{k_1} + \frac{b_2}{k_2} + \frac{b_1}{k_2} = \frac{T_1}{\beta} + \beta T_2 \quad (\beta > 1)$$

Then $X_o(s)/X_i(s)$ can be simplified as

$$\frac{X_o(s)}{X_i(s)} = \frac{(T_1s + 1)(T_2s + 1)}{\left(\frac{T_1}{\beta}s + 1\right)(\beta T_2s + 1)} = \frac{\left(s + \frac{1}{T_1}\right)\left(s + \frac{1}{T_2}\right)}{\left(s + \frac{\beta}{T_1}\right)\left(s + \frac{1}{\beta T_2}\right)}$$

From this transfer function we see that this mechanical system is a mechanical lag-lead network.

A-7-3. Consider the electrical network shown in Figure 7-27. Derive the transfer function of the network. (As usual in the derivation of the transfer function of any four-terminal network, we

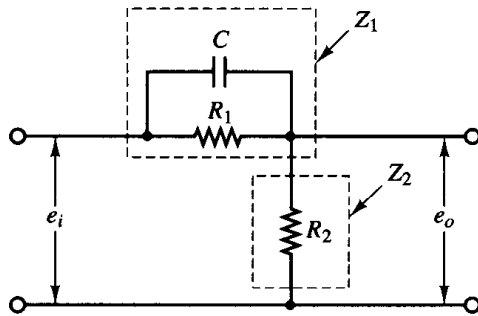


Figure 7-27
Electrical network.

assume that the source impedance that the network sees is zero and that the output load impedance is infinite.)

Solution. Using the symbols defined in Figure 7-27, we find that the complex impedances Z_1 and Z_2 are

$$Z_1 = \frac{R_1}{R_1 C s + 1}, \quad Z_2 = R_2$$

The transfer function between the output $E_o(s)$ and the input $E_i(s)$ is

$$\frac{E_o(s)}{E_i(s)} = \frac{Z_2}{Z_1 + Z_2} = \frac{R_2}{R_1 + R_2} \frac{R_1 C s + 1}{\frac{R_1 R_2}{R_1 + R_2} C s + 1}$$

Define

$$R_1 C = T, \quad \frac{R_2}{R_1 + R_2} = \alpha < 1$$

Then the transfer function becomes

$$\frac{E_o(s)}{E_i(s)} = \alpha \frac{T s + 1}{\alpha T s + 1} = \frac{s + \frac{1}{T}}{s + \frac{1}{\alpha T}}$$

Since α is less than 1, this network is a lead network.

A-7-4. Obtain the transfer function of the network shown in Figure 7-28.

Solution. The complex impedances Z_1 and Z_2 are

$$Z_1 = \frac{R_1}{R_1 C_1 s + 1}, \quad Z_2 = R_2 + \frac{1}{C_2 s}$$

The transfer function between $E_o(s)$ and $E_i(s)$ is

$$\frac{E_o(s)}{E_i(s)} = \frac{Z_2}{Z_1 + Z_2} = \frac{(R_1 C_1 s + 1)(R_2 C_2 s + 1)}{(R_1 C_1 s + 1)(R_2 C_2 s + 1) + R_1 C_2 s}$$

The denominator of this transfer function can be factored into two real terms. Let us define

$$R_1 C_1 = T_1, \quad R_2 C_2 = T_2, \quad R_1 C_1 + R_2 C_2 + R_1 C_2 = \frac{T_1}{\beta} + \beta T_2 \quad (\beta > 1)$$

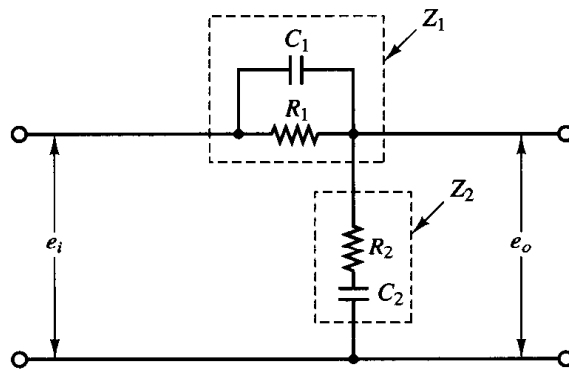


Figure 7-28
Electrical network.

Then $E_o(s)/E_i(s)$ can be simplified to

$$\frac{E_o(s)}{E_i(s)} = \frac{(T_1s + 1)(T_2s + 1)}{\left(\frac{T_1}{\beta}s + 1\right)(\beta T_2s + 1)} = \frac{\left(s + \frac{1}{T_1}\right)\left(s + \frac{1}{T_2}\right)}{\left(s + \frac{\beta}{T_1}\right)\left(s + \frac{1}{\beta T_2}\right)}$$

This is a lag-lead network.

A-7-5. A control system with

$$G(s) = \frac{K}{s^2(s + 1)}, \quad H(s) = 1$$

is unstable for all positive values of gain K .

Plot the root loci of the system. By using this plot, show that this system can be stabilized by adding a zero on the negative real axis or by modifying $G(s)$ to $G_1(s)$, where

$$G_1(s) = \frac{K(s + \alpha)}{s^2(s + 1)} \quad (0 \leq \alpha < 1)$$

Solution. A root-locus plot for the system with

$$G(s) = \frac{K}{s^2(s + 1)}, \quad H(s) = 1$$

is shown in Figure 7-29(a). Since two branches lie in the right half-plane, the system is unstable for any value of $K > 0$.

Addition of a zero to the transfer function $G(s)$ bends the right half-plane branches to the left and brings all root-locus branches to the left half-plane, as shown in the root-locus plot in Figure 7-29(b). Thus, the system with

$$G_1(s) = \frac{K(s + \alpha)}{s^2(s + 1)}, \quad H(s) = 1 \quad (0 \leq \alpha < 1)$$

is stable for all $K > 0$.

A-7-6. Consider a system with an unstable plant as shown in Figure 7-30a. Using the root-locus approach, design a proportional-plus-derivative controller (that is, determine the values of K_p and T_d) such that the damping ratio ζ of the closed-loop system is 0.7 and the undamped natural frequency ω_n is 0.5 rad/sec.

Solution. Note that the open-loop transfer function involves two poles at $s = 1.085$ and $s = -1.085$ and one zero at $s = -1/T_d$, which is unknown at this point.

Since the desired closed-loop poles must have $\omega_n = 0.5$ rad/sec and $\zeta = 0.7$, they must be located at

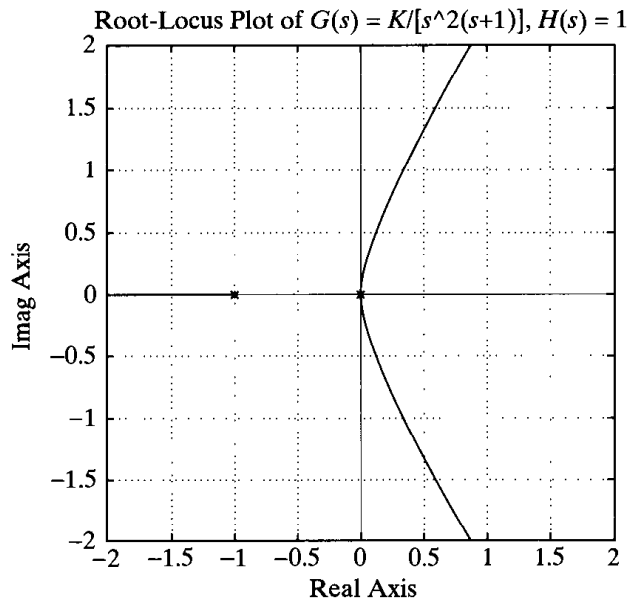
$$s = 0.5 \angle 180^\circ \pm 45.573^\circ$$

($\zeta = 0.7$ corresponds to a line having an angle of 45.573° with the negative real axis.) Hence, the desired closed-loop poles are at

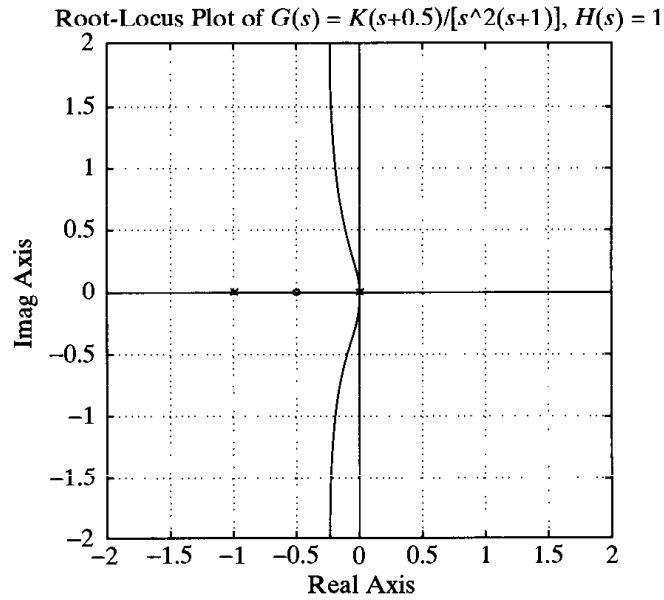
$$s = -0.35 \pm j0.357$$

The open-loop poles and the desired closed-loop pole in the upper half-plane are located in the diagram shown in Figure 7-30b. The angle deficiency at point $s = -0.35 + j0.357$ is

$$-166.026^\circ - 25.913^\circ + 180^\circ = -11.938^\circ$$



(a)



(b)

Figure 7-29

(a) Root-locus plot of the system with $G(s) = K/[s^2(s+1)]$ and $H(s) = 1$;
 (b) root-locus plot of the system with $G_1(s) = K(s+a)/[s^2(s+1)]$ and $H(s) = 1$, where $a = 0.5$.

This means that the zero at $s = -1/T_d$ must contribute 11.938° , which, in turn, determines the location of the zero as follows:

$$s = -\frac{1}{T_d} = -2.039$$

Hence, we have

$$K_p(1 + T_d s) = K_p T_d \left(\frac{1}{T_d} + s \right) = K_p T_d (s + 2.039) \quad (7-7)$$

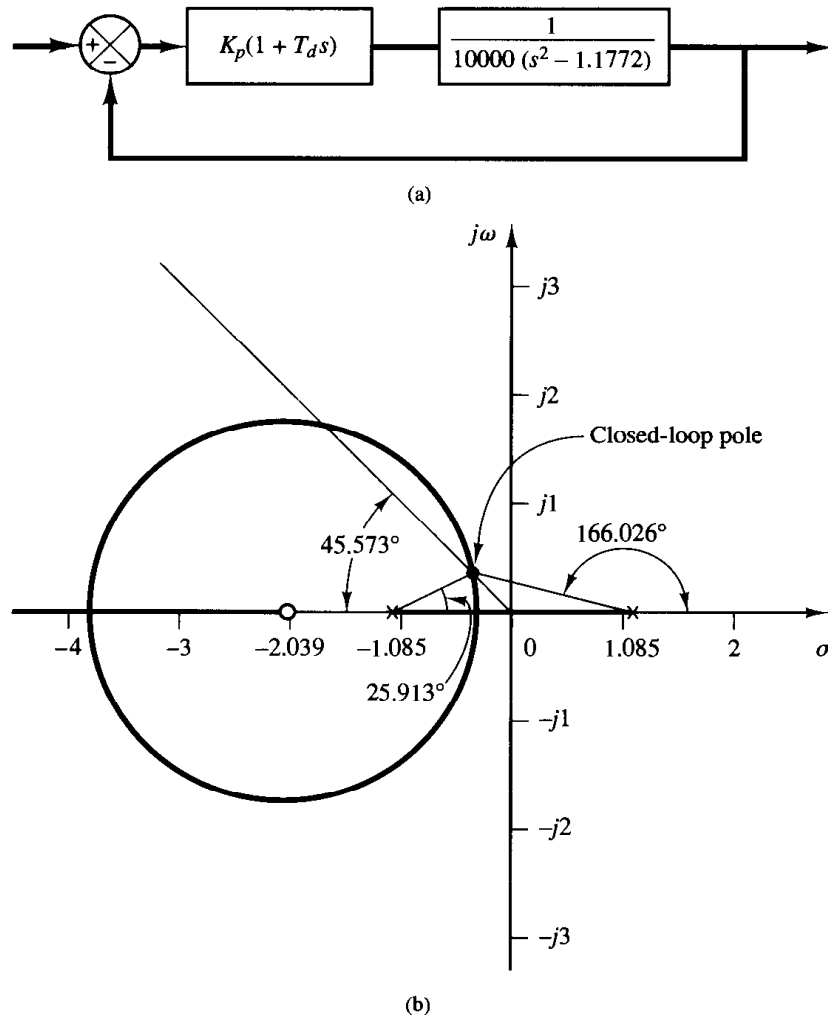


Figure 7-30
(a) PD control of an unstable plant; (b) root-locus diagram for the system.

The value of T_d is

$$T_d = \frac{1}{2.039} = 0.4904$$

The value of gain K_p can be determined from the magnitude condition as follows:

$$\left| K_p T_d \frac{s + 2.039}{10000(s^2 - 1.1772)} \right|_{s=-0.35+j0.357} = 1$$

or

$$K_p T_d = 6999.5$$

Hence,

$$K_p = \frac{6999.5}{0.4904} = 14,273$$

By substituting the numerical values of T_d and K_p into Equation (7-7), we obtain

$$K_p(1 + T_d s) = 14,273(1 + 0.4904s) = 6999.5(s + 2.039)$$

which gives the transfer function of the desired proportional-plus-derivative controller.

A-7-7. Consider the control system shown in Figure 7–31. Design a lag compensator $G_c(s)$ such that the static velocity error constant K_v is 50 sec^{-1} without appreciably changing the location of the original closed-loop poles, which are at $s = -2 \pm j\sqrt{6}$.

Solution. Assume that the transfer function of the lag compensator is

$$G_c(s) = \hat{K}_c \frac{s + \frac{1}{T}}{s + \frac{1}{\beta T}} \quad (\beta > 1)$$

Since K_v is specified as 50 sec^{-1} , we have

$$K_v = \lim_{s \rightarrow 0} s G_c(s) \frac{10}{s(s+4)} = \hat{K}_c \beta 2.5 = 50$$

Thus

$$\hat{K}_c \beta = 20$$

Now choose $\hat{K}_c = 1$. Then

$$\beta = 20$$

Choose $T = 10$. Then the lag compensator can be given by

$$G_c(s) = \frac{s + 0.1}{s + 0.005}$$

The angle contribution of the lag compensator at the closed-loop pole $s = -2 + j\sqrt{6}$ is

$$\begin{aligned} \angle G_c(s) \Big|_{s = -2 + j\sqrt{6}} &= \tan^{-1} \frac{\sqrt{6}}{-1.9} - \tan^{-1} \frac{\sqrt{6}}{-1.995} \\ &= -1.3616^\circ \end{aligned}$$

which is small. Thus the change in the location of the dominant closed-loop poles is very small.

The open-loop transfer function of the system becomes

$$G_c(s)G(s) = \frac{s + 0.1}{s + 0.005} \frac{10}{s(s+4)}$$

The closed-loop transfer function is

$$\frac{C(s)}{R(s)} = \frac{10s + 1}{s^3 + 4.005s^2 + 10.02s + 1}$$

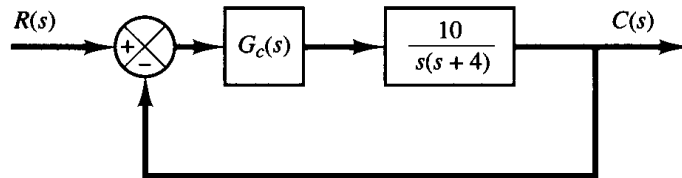


Figure 7–31
Control system.

To compare the transient-response characteristics before and after the compensation, the unit-step and unit-ramp responses of the compensated and uncompensated systems are shown in Figures 7-32(a) and (b), respectively. The steady-state error in the unit-ramp response is shown in Figure 7-32(c).

A-7-8. Consider a unity-feedback control system whose feedforward transfer function is given by

$$G(s) = \frac{10}{s(s + 2)(s + 8)}$$

Design a compensator such that the dominant closed-loop poles are located at $s = -2 \pm j2\sqrt{3}$ and the static velocity error constant K_v is equal to 80 sec^{-1} .

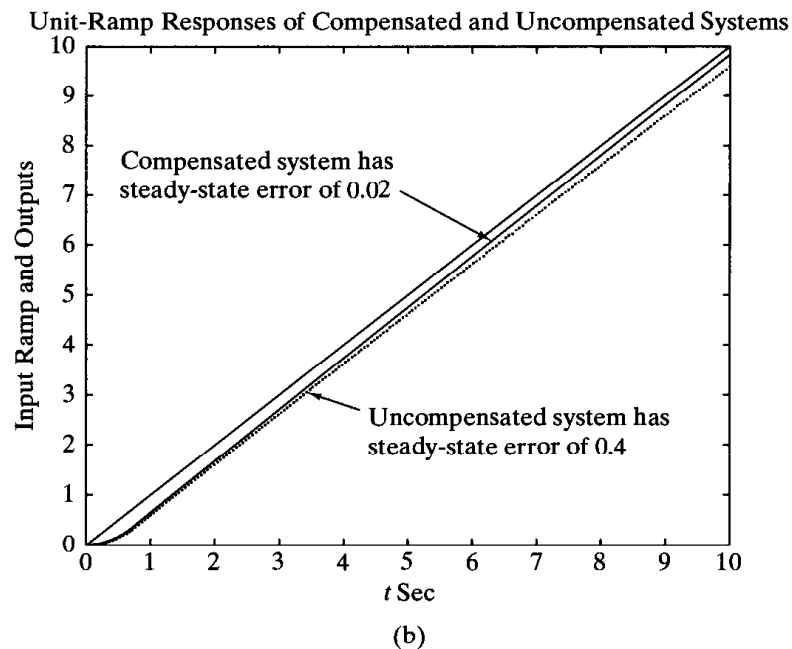
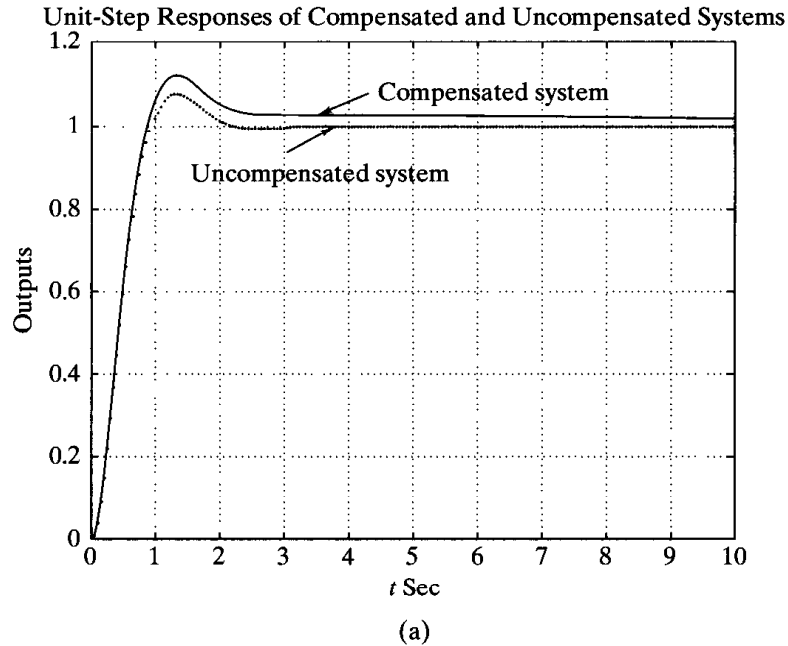


Figure 7-32

(a) Unit-step responses of the compensated and uncompensated systems; (b) unit-ramp responses of both systems; (c) unit-ramp responses showing steady-state errors.

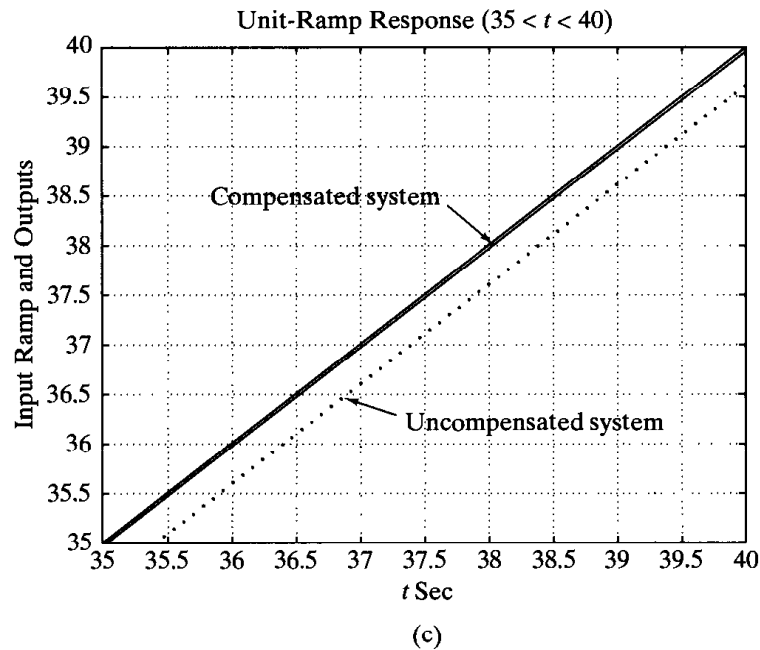


Figure 7-32
(Continued)

Solution. The static velocity error constant of the uncompensated system is $K_v = \frac{10}{16} = 0.625$. Since $K_v = 80$ is required, we need to increase the open-loop gain by 128. (This implies that we need a lag compensator.) The root-locus plot of the uncompensated system reveals that it is not possible to bring the dominant closed-loop poles to $-2 \pm j2\sqrt{3}$ by just a gain adjustment alone. See Figure 7-33. (This means that we also need a lead compensator.) Therefore, we shall employ a lag-lead compensator.

Let us assume that the transfer function of the lag-lead compensator to be

$$G_c(s) = K_c \left(\frac{s + \frac{1}{T_1}}{s + \frac{\beta}{T_1}} \right) \left(\frac{s + \frac{1}{T_2}}{s + \frac{1}{\beta T_2}} \right) \quad (\alpha = \beta)$$

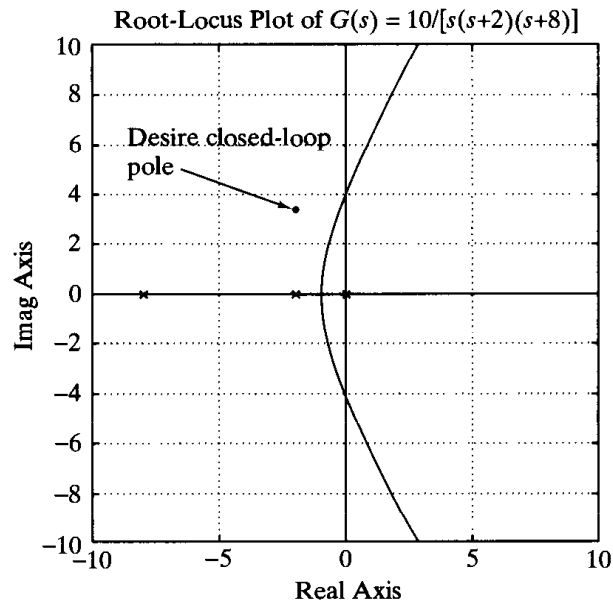


Figure 7-33
Root-locus plot of
 $G(s) = 10/[s(s+2)(s+8)]$.